Introduction to pharmacology

Learning Objectives

- At the end of this chapter the student will be able to:
  - Define various terminologies used in Pharmacology
  - Understand pharmacokinetic parameters like absorption, distribution, metabolism and excretion of drugs.
  - Understand pharmacodynamic principles like mechanism of drug action, dose response relationship.
  - Understand drug safety and effectiveness like factors affecting drug action and adverse drug reactions.
  - Understand new drug development and evaluation.

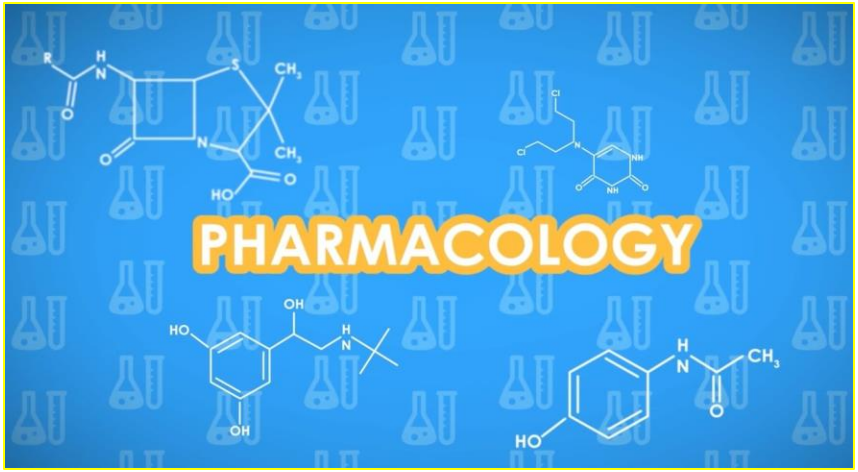
How drugs act?

- Take paracetamol as an example
- How did it relieve a pain?
- How did it reduce fever?



Pharmacology, definition

- Pharmacognosy
- Pharmaceutics
- Pharmacology
- Pharmacotherapy
- Pharmacogenomics
- Pharmacoepidemiology
- Pharmacoeconomics

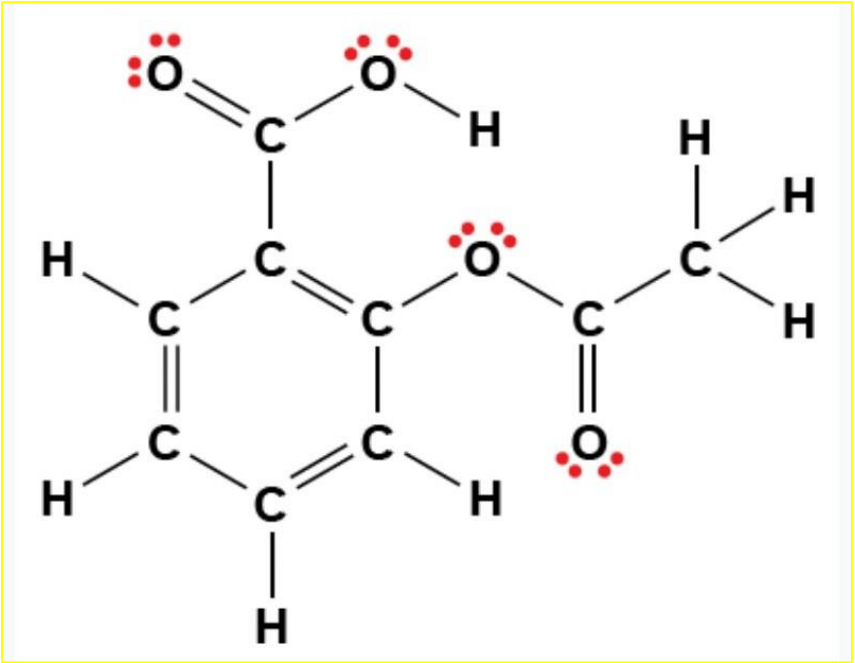


Pharmacon (drug) + *logos* (study)

Study of the effects of drugs on the function of living systems.

Drug

- Any substance/ product used or intended to be used to modify/ explore physiological systems or pathological states for the benefit of the recipient.



Notes about drugs

- Drugs are used to :-
  - Diagnose a disease (e.g. salbutamol)
  - Prevent a disease (e.g. vaccines)
  - Cure a disease (e.g. antibiotics)
  - Control a disease (e.g. HAART)

- FDA – All drugs are chemicals, BUT, all chemicals are not drugs; All drugs are poisons, BUT, all poisons are not drugs.

Drug vs. Medicine

Drug



Medicine



Medicine = Drug + additives

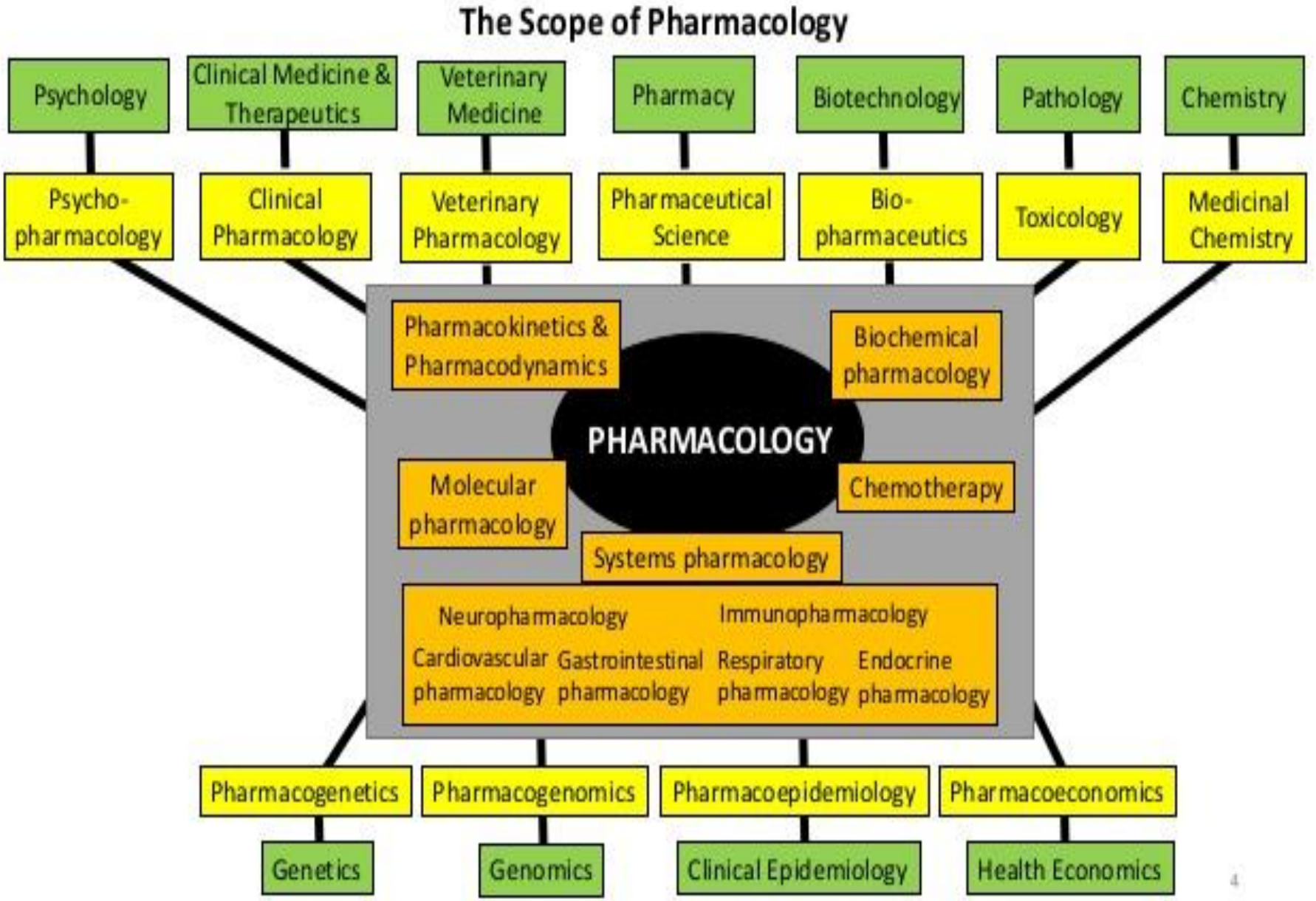
Drug Vs. Poison

- Poisons – Substances that cause harmful, dangerous or fatal symptoms in living substances.

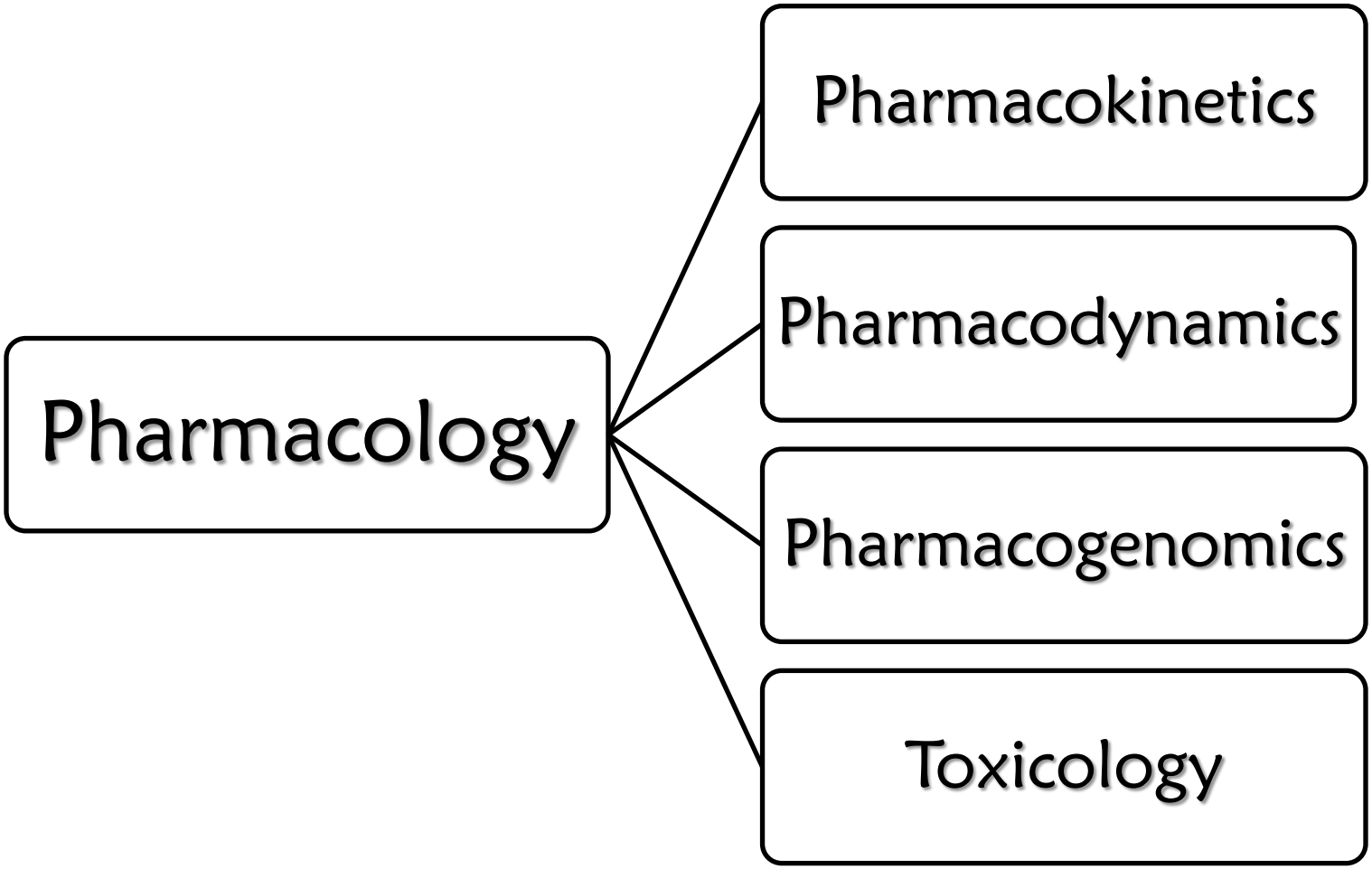
Poisons – toxic at low doses

Drugs – toxic at higher doses

All drugs are poisons; what matters is the dose!
(Paracelsus; 1493 – 1541)


The Scope of Pharmacology
Psychology
Clinical Medicine & Therapeutics
Veterinary Medicine
Pharmacy
Biotechnology
Pathology
Chemistry
Psycho-pharmacology
Clinical Pharmacology
Veterinary Pharmacology
Pharmaceutical Science
Bio-pharmaceutics
Toxicology
Medicinal Chemistry
Pharmacokinetics & Pharmacodynamics
Biochemical pharmacology
PHARMACOLOGY
Molecular pharmacology
Chemotherapy
Systems pharmacology
Neuropharmacology
Immunopharmacology
Cardiovascular pharmacology
Gastrointestinal pharmacology
Respiratory pharmacology
Endocrine pharmacology
Pharmacogenetics
Pharmacogenomics
Pharmacoepidemiology
Pharmacoeconomics
Genetics
Genomics
Clinical Epidemiology
Health Economics

Branches of pharmacology



History of pharmacology

- Prehistoric people recognized the beneficial or toxic effects of many plant and animal materials.

- Records from China, Egypt and India list remedies of many types including a few that are still recognized as useful drugs today.

- Most were worthless or actually harmful.

Ancient philosophies

- Allopathy
  - Blood letting
  - Use of emetics and cathartics
- Homeopathy
  - ✓Like cures the like
- Doctrine of signature

Allopathy

James Gregory

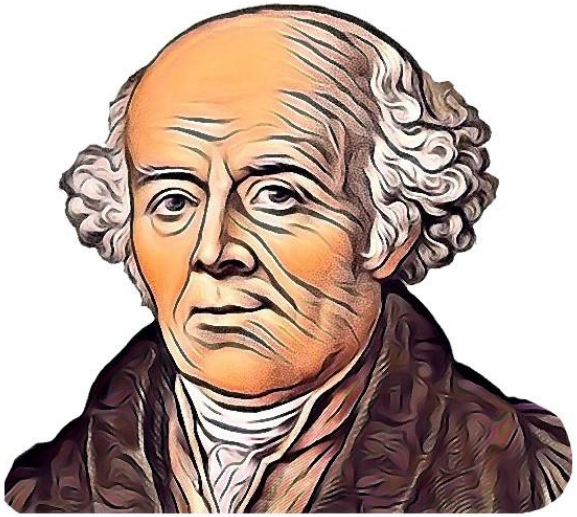


Blood letting

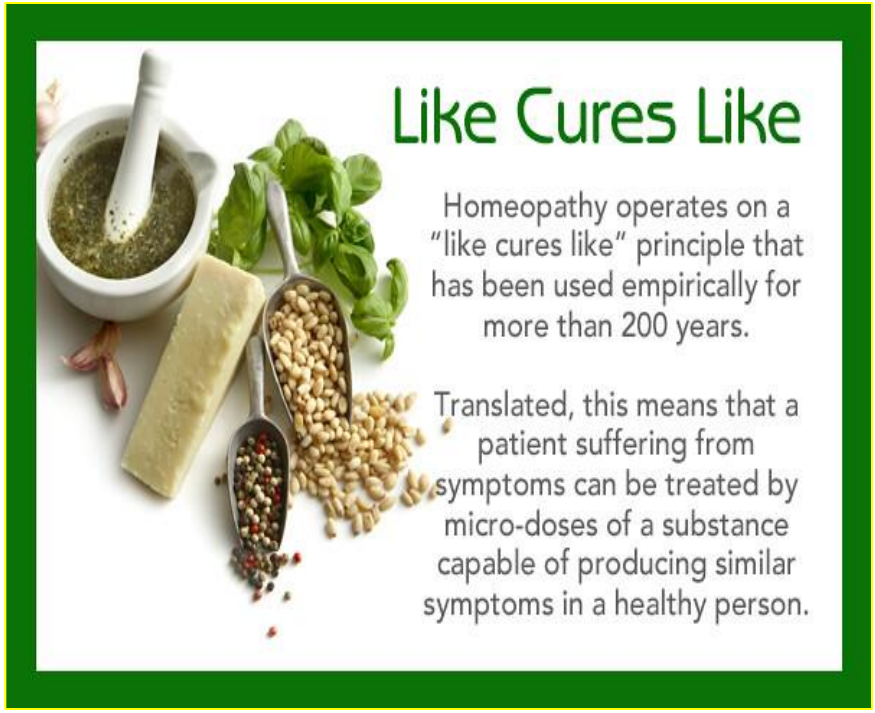


Homeopathy

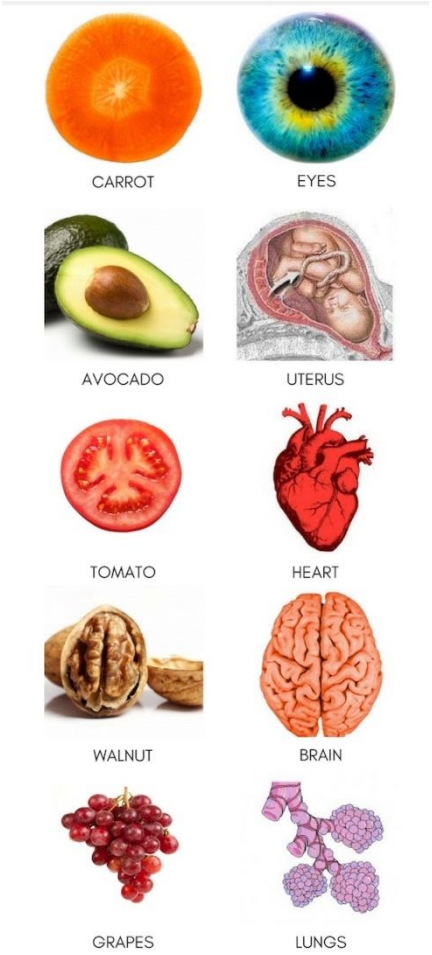
Samuel Hahnemann

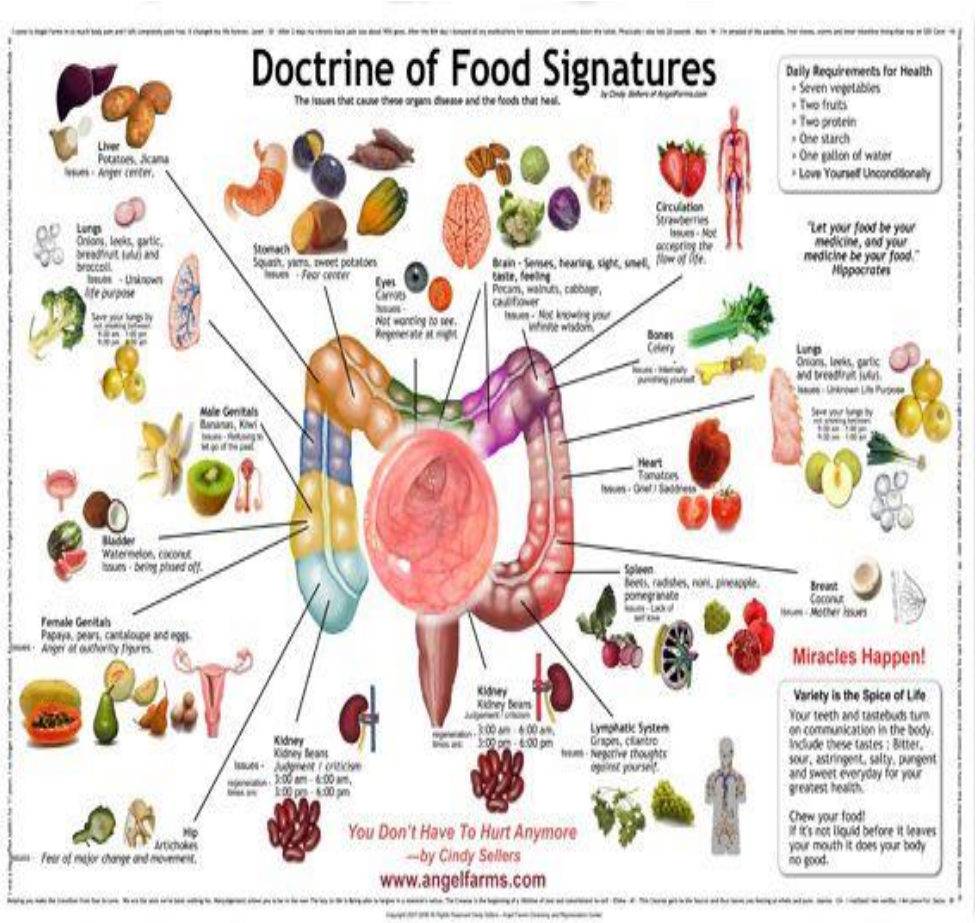


Like cures like



Doctrine of signature





Modern pharmacology

Rudolf Buchheim



Claude Bernard



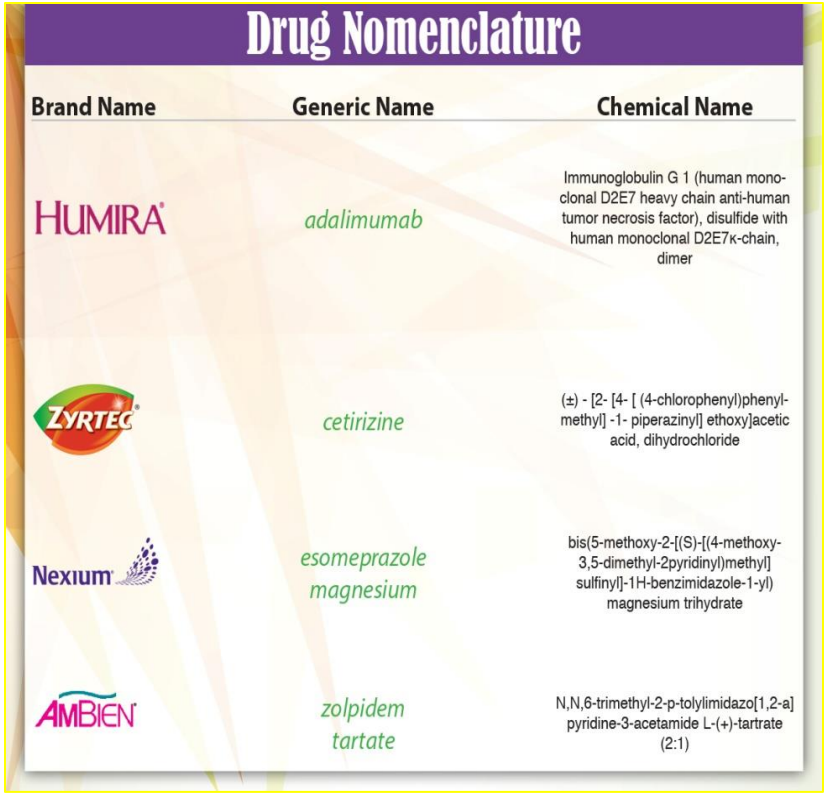
Rudolph – created the first pharmacology institute at his home
Claude – developed methods of experimental physiology and pharmacology.

SOURCES OF DRUGS

- Natural sources
  - Plants sources
  - Animals sources
  - Microbiological sources.
  - Mineral sources. e.g.-$FeSO_4$, $MgSO_4$
  - Genetically engineered drugs.e.g.-Hepatitis B vaccine
- Synthetic/semi-synthetic sources e.g.- Aspirin,Paracetamol

Drug names

1. Chemical name
2. Brand name
   - ✓ Proprietary name
   - ✓ Trade name
3. Generic name
   - ✓ Non proprietary name



FDA drug categories

Prescription drugs



Over the counter (OTC) drugs

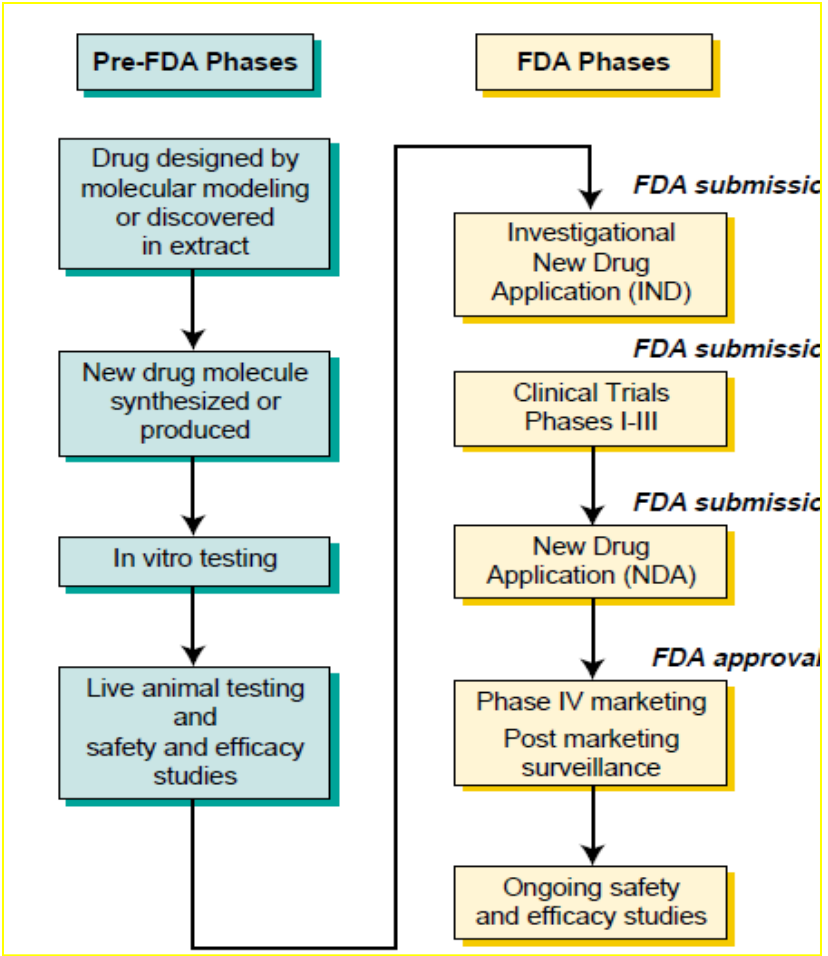


Controlled substances



Drug development

- ✓ How long does it take (in average) for a drug to be approved for use?
  - a) 3 - 5 years
  - b) 5 - 7 years
  - c) 10 - 15 years
  - d) > 20 years



Drug development

- There are 2 stages
  - Preclinical stage
    - In vitro
      - Cell culture
      - Isolated organ
    - In vivo
      - Live animals
  - Clinical stage
    - Human subjects





Live animals testing

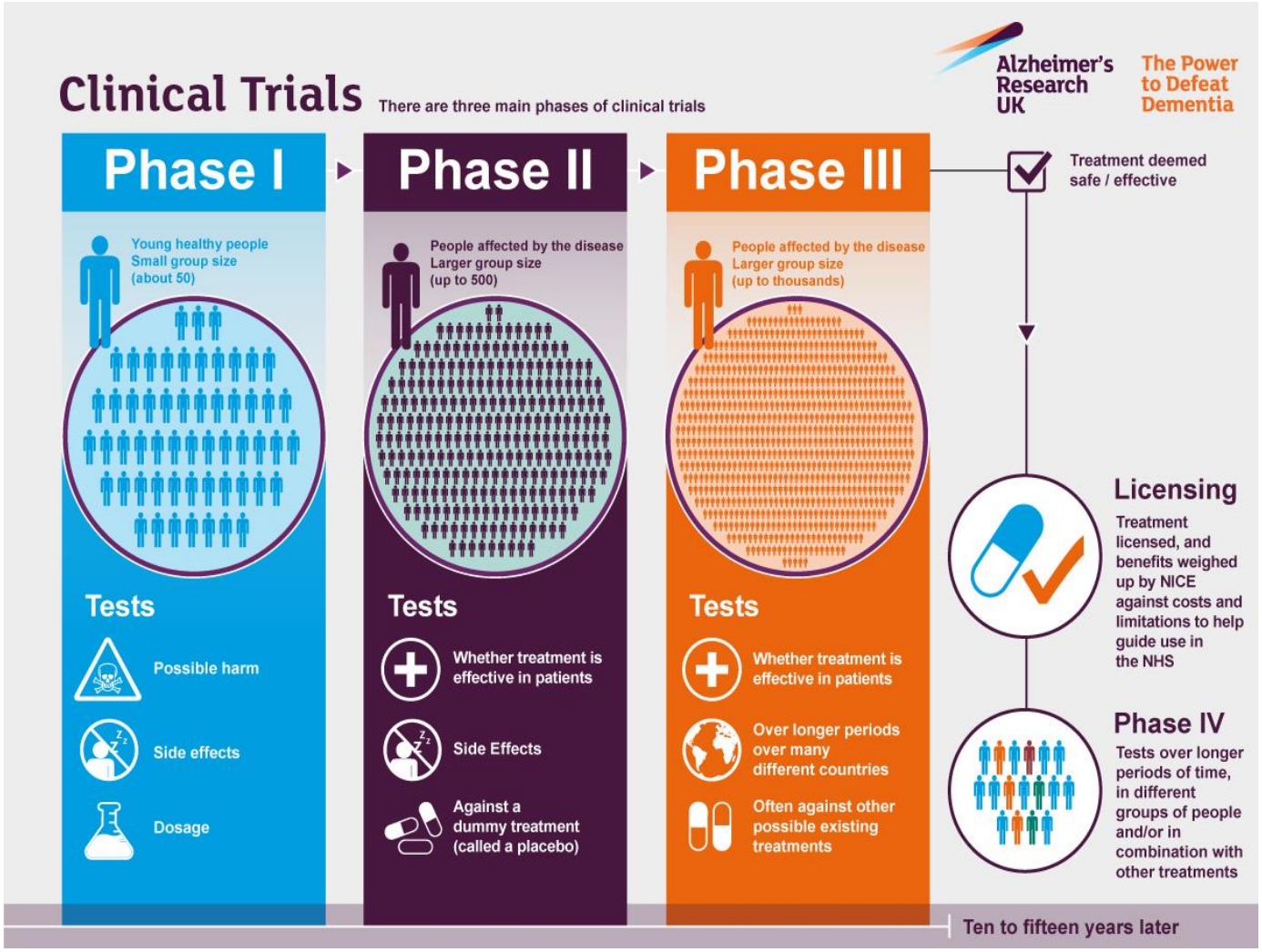








Test in human (clinical trial)



Exceptions in clinical trials

- ✓ Pregnant women
- ✓ Nursing mothers
- ✓ Children
- ✓ Elderly



Pharmacokinetics and pharmacodynamics

Drug activity within the body

1. Pharmaceutic phase
2. Pharmacokinetic phase
3. Pharmacodynamic phase

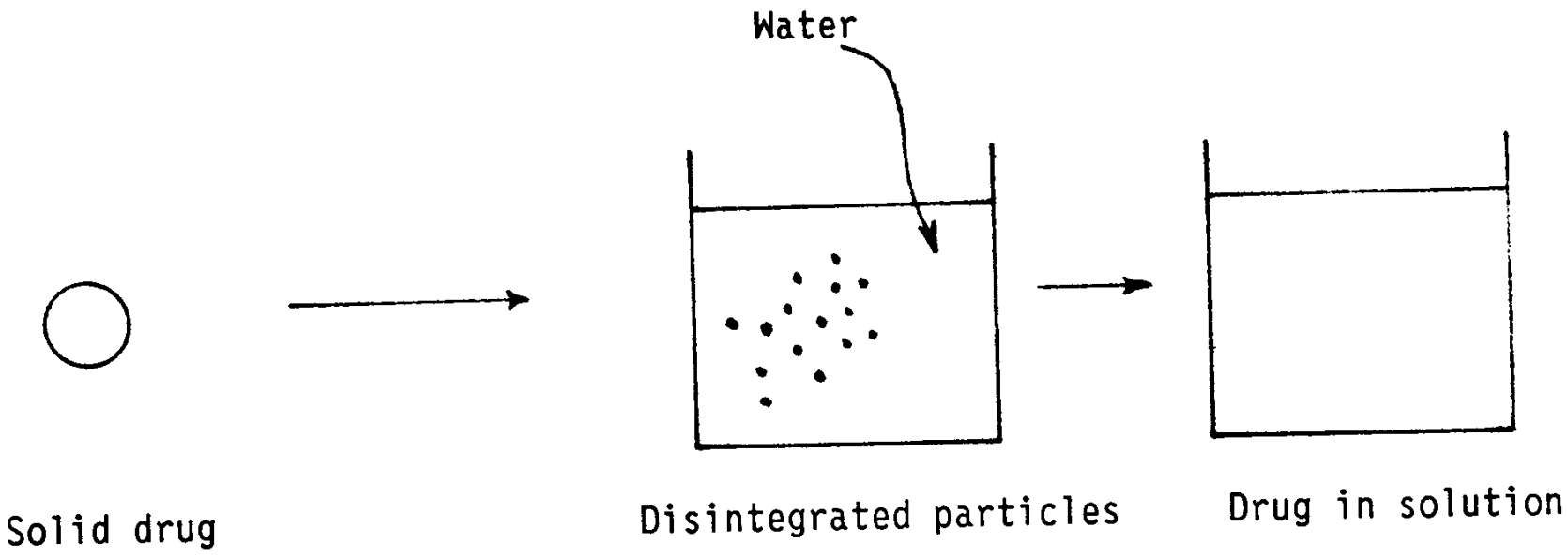






- ❑ Pharmaceutic phase
  - ✓Disintegration
  - ✓Dissolution

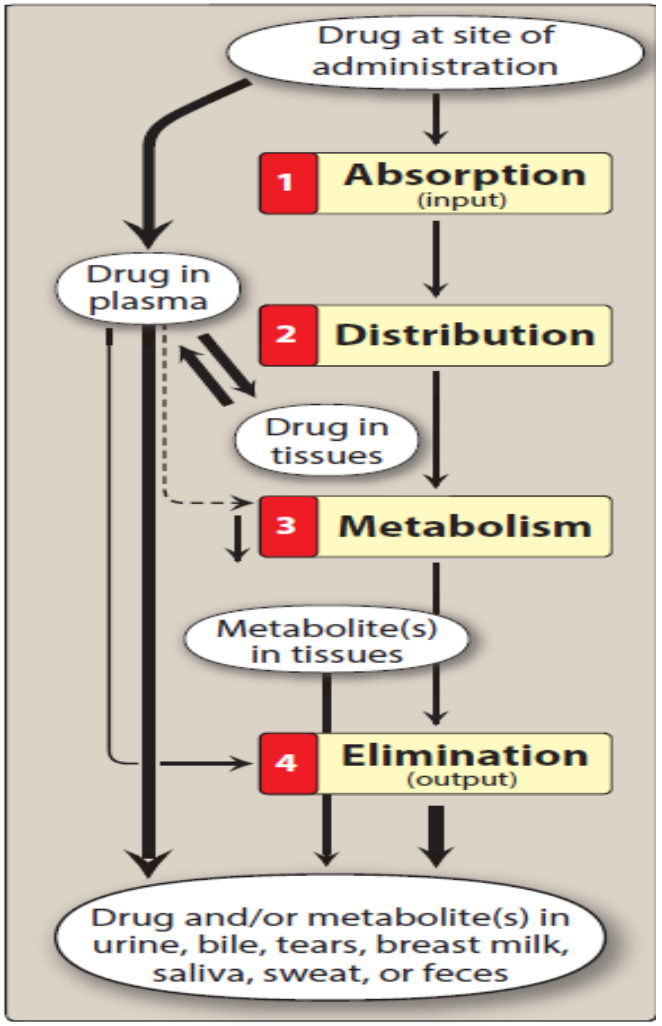
Pharmaceutic phase



Note : Drugs that are liquid or drugs given by injection do not go through the pharmaceutic phase

Pharmacokinetic phase

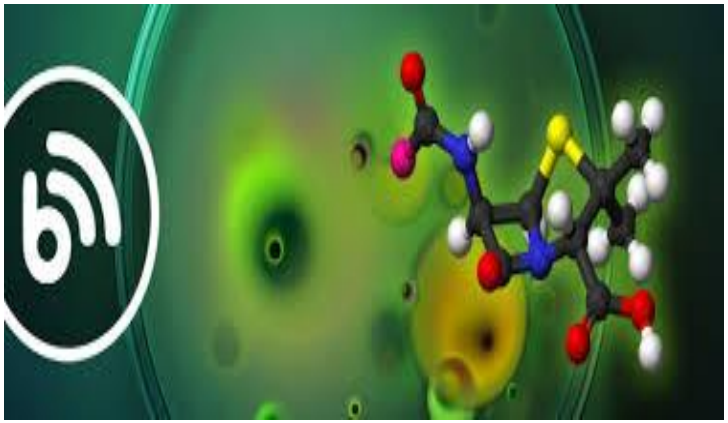
- *Pharmacon* (drug) + *kinetics* (movement)
- Movement of drug into, through, and out of the body.
- What the body does to a drug.

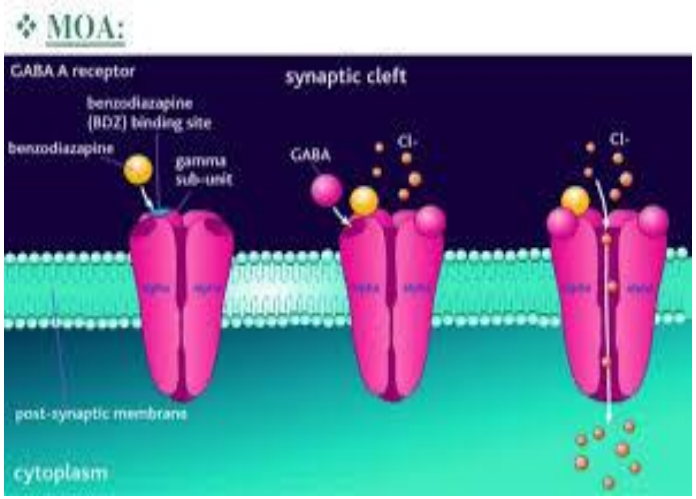


Acronym - ADME

Pharmacodynamic phase

- *Pharmacon* (drug) + *dynamikos* (force, power).
- Deals with the drug's action and effect within the body (Mechanism of action).
- What a drug does to the body.





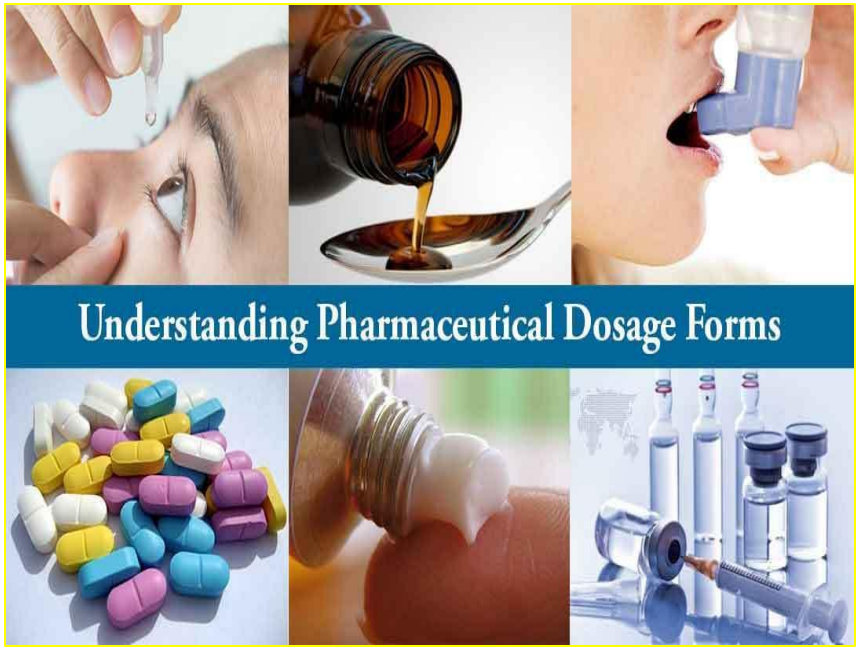
1. Absorption

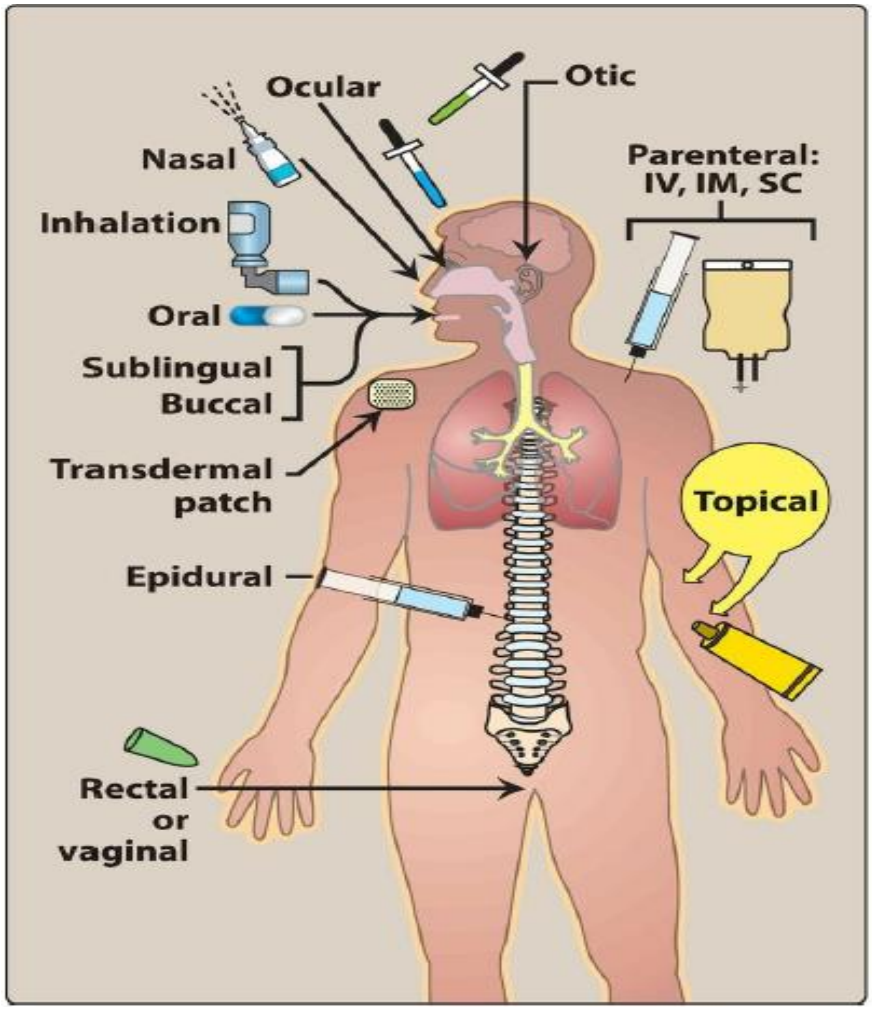
- A process by which a drug enters into a systemic circulation from the site of administration through biological barrier.

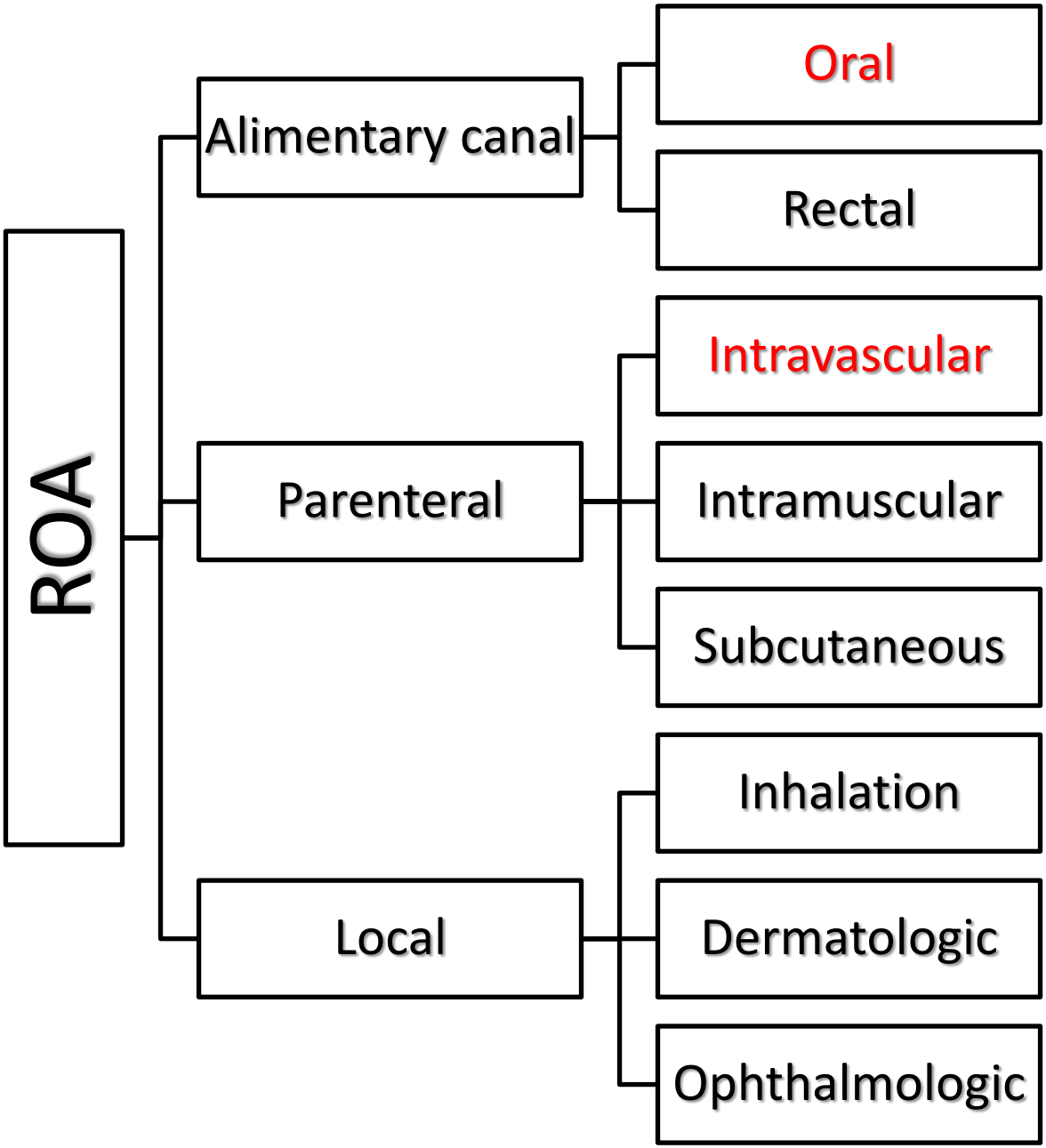
- For a drug to be absorbed, it should first be delivered from an appropriate route of administration (ROA).

- ROA
  - Sites of the body by which a drug is administered.

ROA ….. Why diversified?






ROA
Alimentary canal
Oral
Rectal
Parenteral
Intravascular
Intramuscular
Subcutaneous
Local
Inhalation
Dermatologic
Ophthalmologic

Alimentary canal



Oral administration

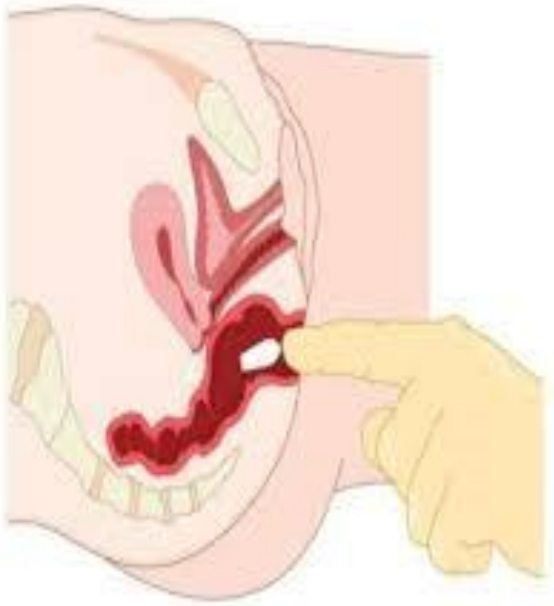


Sublingual administration is where the dosage form is placed under the tongue

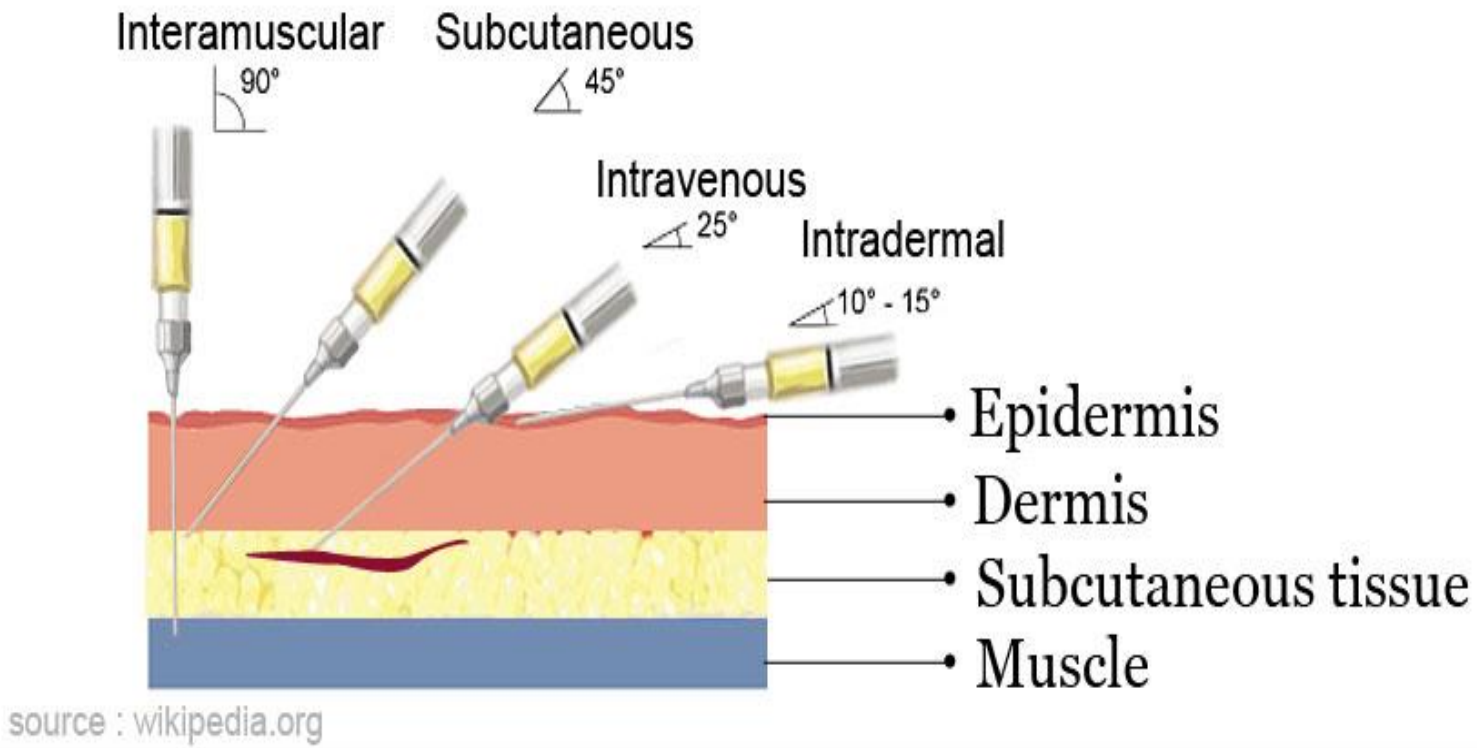


Buccal administration is where the dosage form is placed between gums and inner lining of the cheek

Rectal



Types of Parenteral Routes of Administration



Order of drug absorption

IV > IM > SC

NB : In an I.V injection, absorption is 100%.

Local



Inhalation - sprays or powders for lungs

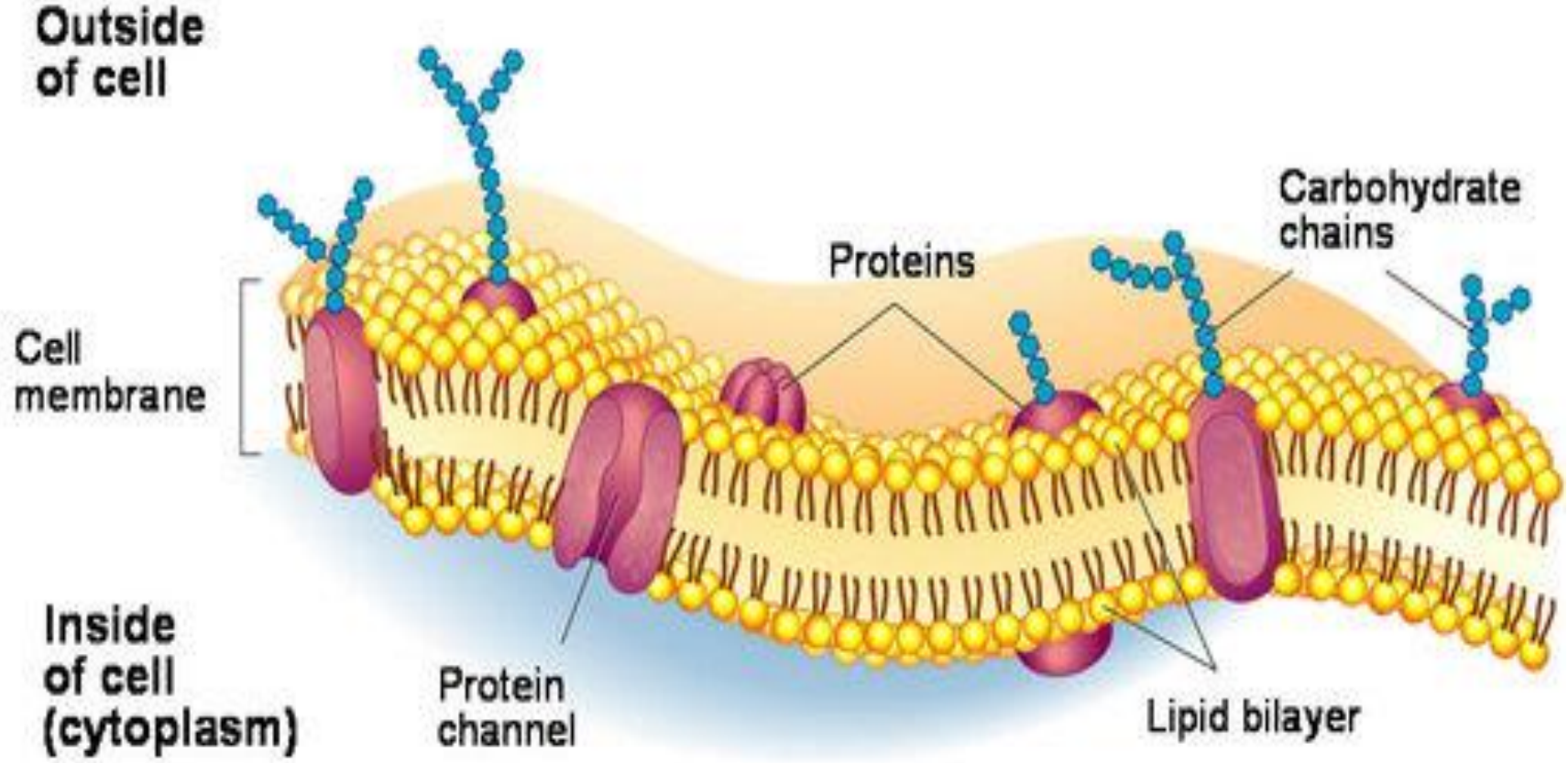


Topical administration: e.g. creams, ointments, lotions for skin; solutions, suspensions, ointments for ear or eye;

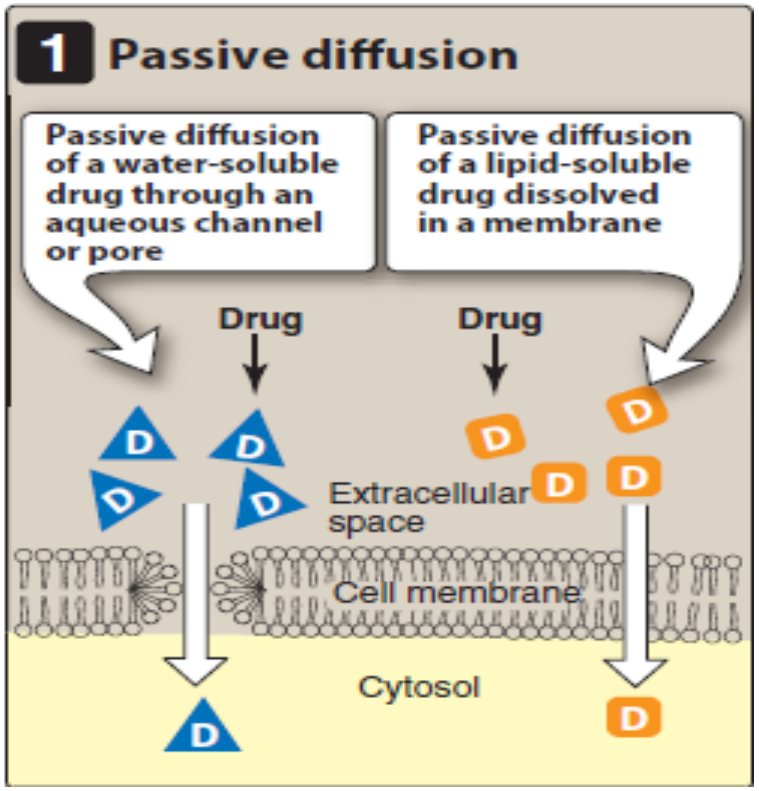
ROA … pros and cons

Route	Advantage	Disadvantage
Oral (PO)	Convenient, easy to administer Large surface area for absorption Inexpensive relative to parenteral administration	Incomplete absorption Hepatic first-pass effect Potential GI irritation
Buccal/Sublingual (SL)	Rapid onset of action No hepatic first-pass effect	Must be lipid-soluble, non-irritating Short duration of action
Rectal (PR)	Almost no hepatic first-pass effect Use when NPO, vomiting, or unconscious	Inconvenient, irritation at site of application Erratic absorption
Intravenous (IV)	No hepatic first-pass effect Slow infusion or rapid onset of action Easy to titrate dose	Hard to remove once administered Risk of infection, bleeding, vascular injury extravasation Expensive
Intramuscular (IM)	Depot storage if oil-based = slow release of drug Aqueous solution = rapid onset of action	Pain/hematoma at site of injection
Subcutaneous (SC)	Non-irritating drugs, small volumes Constant, even absorption Alternative to IV	Pain at site of injection Smaller volumes than IM May have tissue damage from multiple injections
Intrathecal	Direct into CSF Bypass BBB and blood-CSF barrier	Risk of infection
Inhalation	Immediate action in lungs Rapid delivery to blood No hepatic first-pass effect	Must be gas, vapour, or aerosol
Topical	Easy to administer Localized (limited systemic absorption)	Effects are mainly limited to site of application
Transdermal	Drug absorption through intact skin No hepatic first-pass effect	Irritation at site of application Delayed onset of action Hydrophilic drugs not easily absorbed
Others (intraperitoneal, intra-articular)	Local effect	Risk of infection

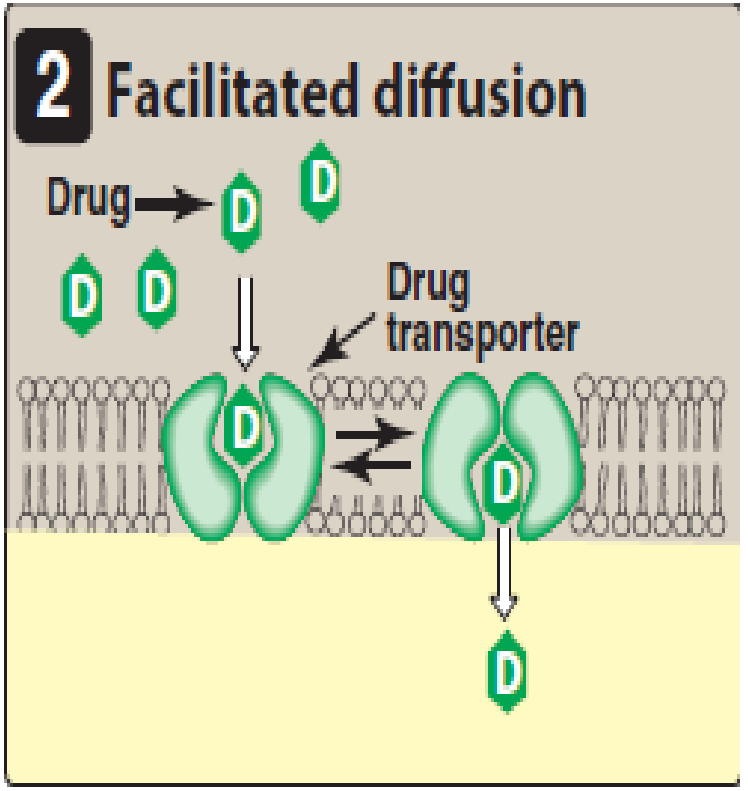
Drug transport : cell membrane



Drug transport across membranes

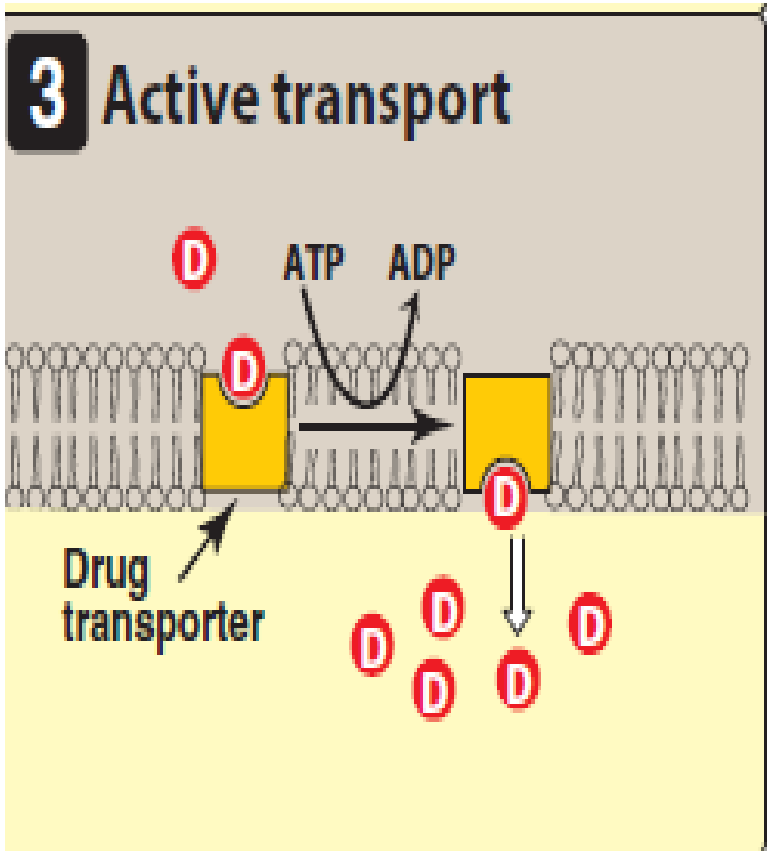


The most common (99%)

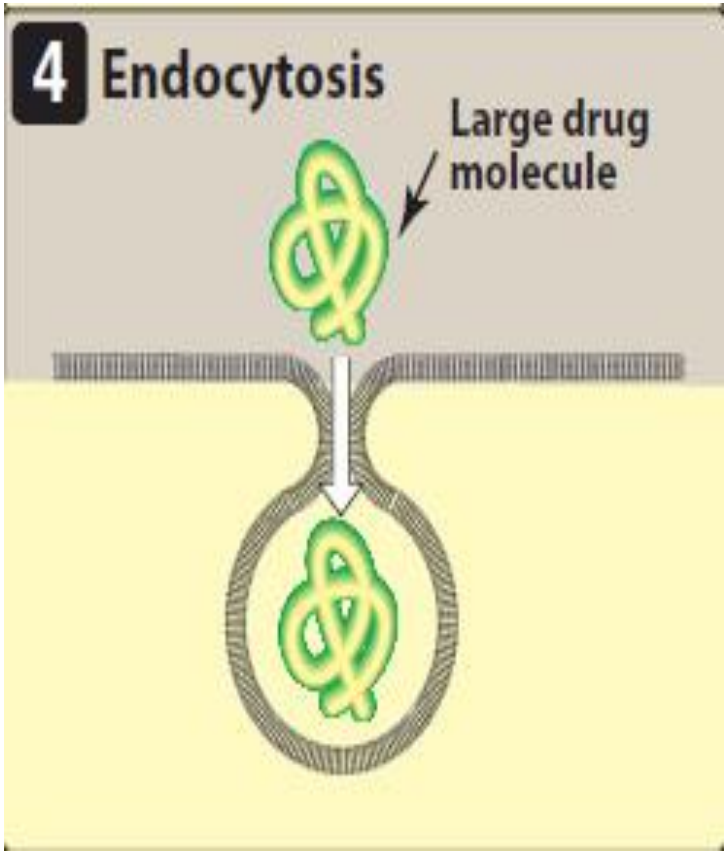


Example – Tetracycline

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Example – levodopa



Example – Vitamin B12

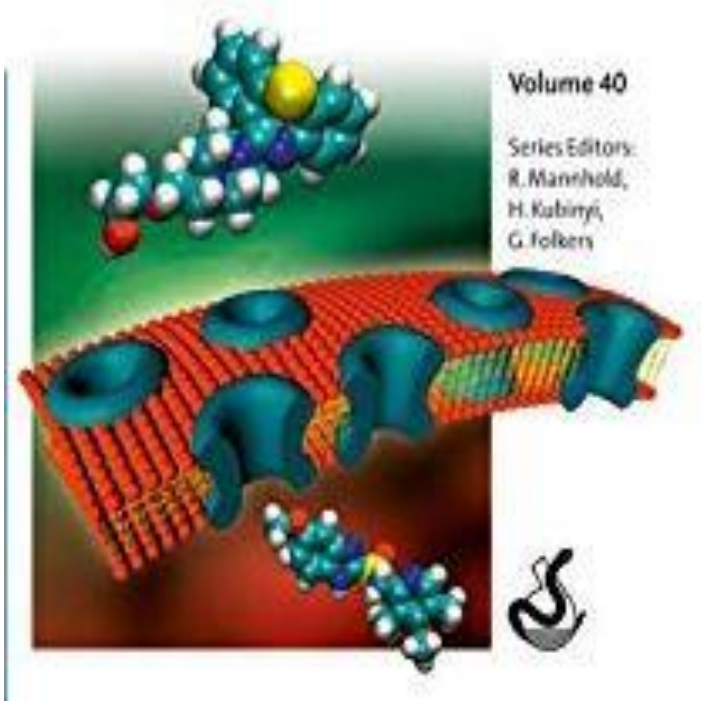
Factors affecting drug absorption

- Blood flow
- Surface area
- Food in GIT
- Stomach acidity

Bioavailability (F)

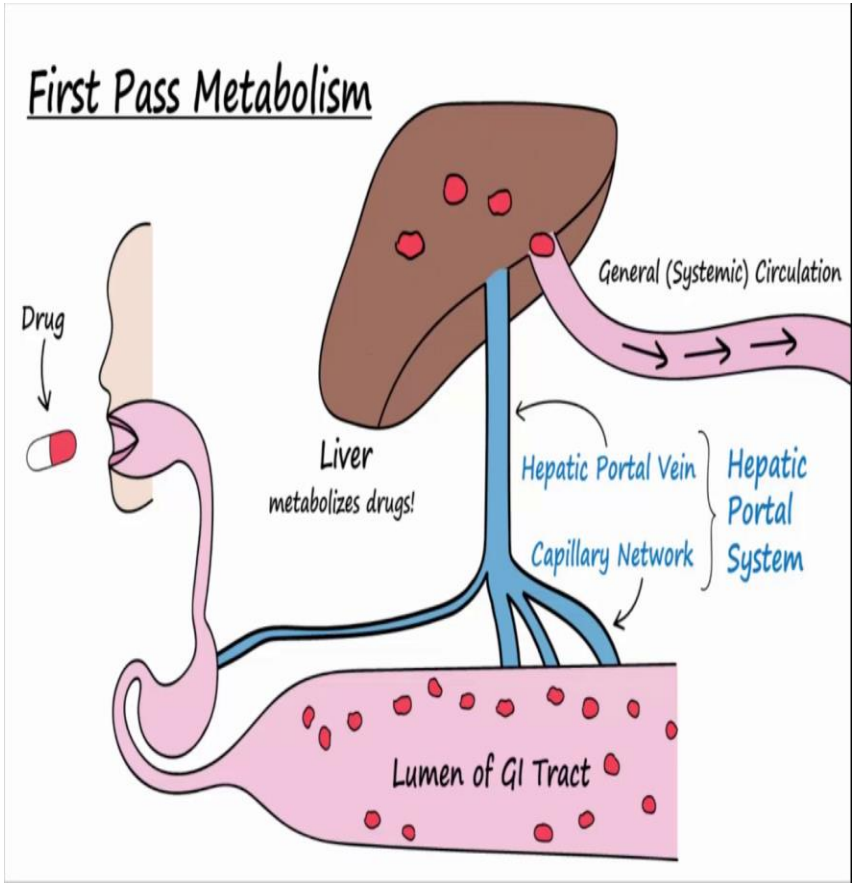
- Proportion of dose that reaches systemic circulation in an unchanged state.
- Decreased by limited drug absorption or first-pass effect.
- When the drug is given IV, the bioavailability is

  100%



First pass hepatic metabolism

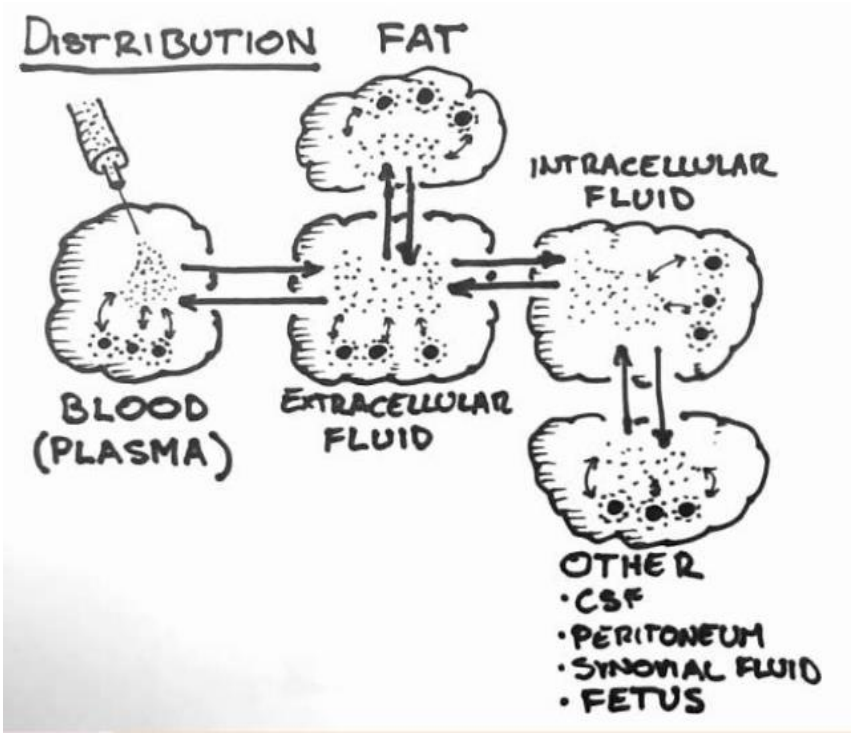
- Drug metabolism by the liver and/or the gut before it reaches systemic circulation.
- Occurs with oral administration of drugs.
- Occurs less with rectal administration.



2. Distribution

- Process by which a drug reversibly leaves the blood-stream and enters the interstitium (extracellular fluid) and the tissues.

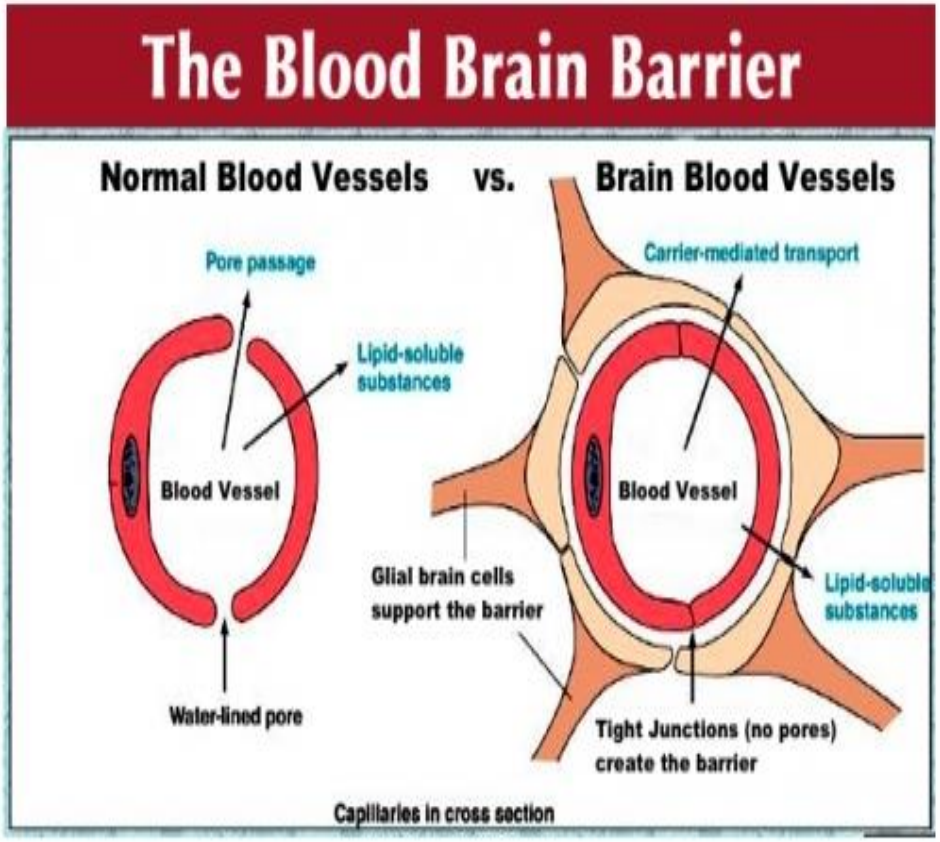
- Movement of drugs between different body compartment and to the site of action.



Factors affecting drug distribution

1. Blood flow
   - Better distribution to brain, liver, and kidney
2. Barriers
   - Blood brain barrier (BBB)
   - Blood placenta barrier (BPB)
   - Blood testes barrier (BTB)
3. Drug binding to plasma proteins and tissues
   - Plasma proteins ….. albumin
   - Tissue binding …… fat tissue, bone tissue

Barriers to distribution



- For a drug to cross BBB, it should be
  - ✓ Lipid soluble
  - ✓ Small in size

e.g. benzodiazepines for anxiety and convulsion

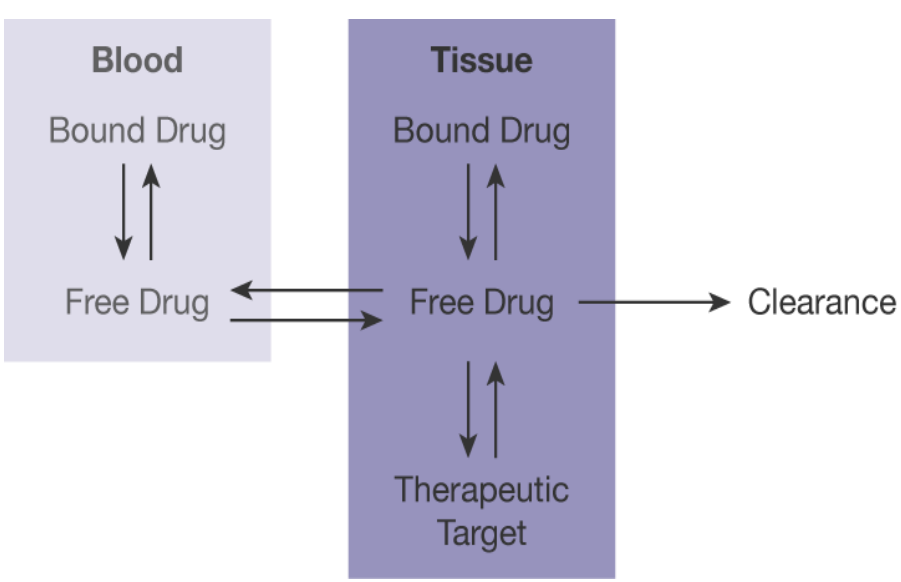
Plasma protein binding

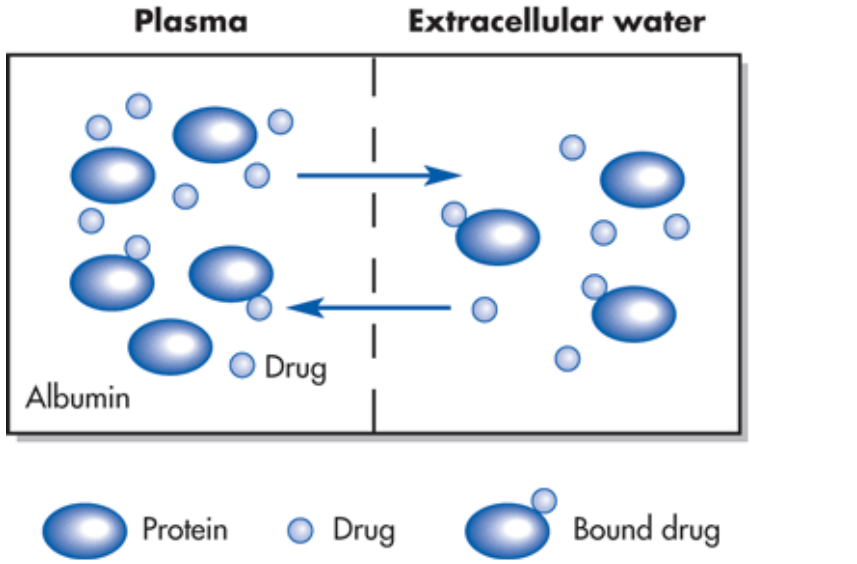
- Drug molecules in the blood exist in an equilibrium of two forms :-

1. Bound to plasma protein: acidic drugs bind to albumin, basic drugs bind to α1-acid glycoprotein

2. Free or unbound: drug can leave the circulation to distribute into tissues and exert an effect, subject to metabolism and elimination.

Binding to plasma proteins





Source: Shargel L, Wu-Pong S, Yu ABC: *Applied Biopharmaceutics & Pharmacokinetics, 6th Edition*: www.accesspharmacy.com

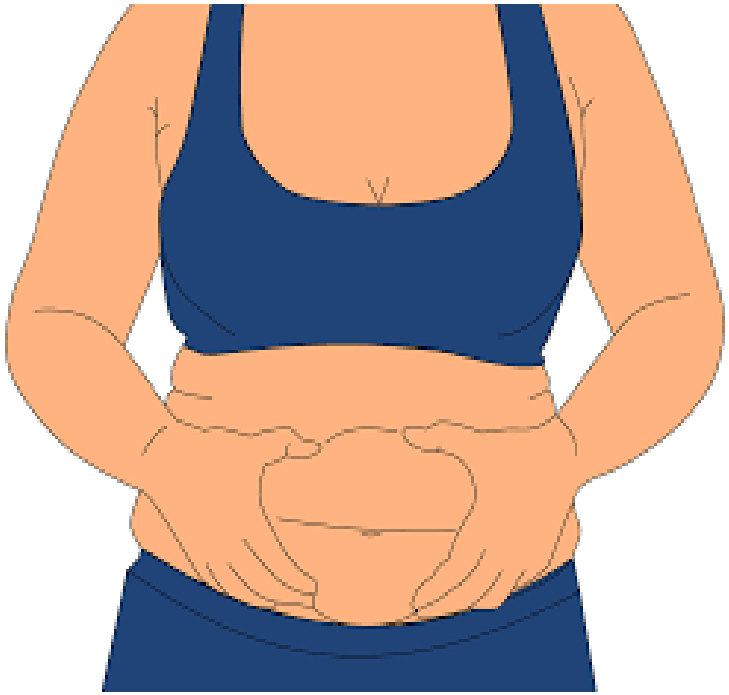
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Binding to tissues

- Many drugs accumulate in tissues, leading to higher concentrations in tissues than in the extracellular fluid and blood.

- Tissue reservoirs may serve as a major source of the drug and prolong its actions or cause local drug toxicity.

Tissue binding … examples

Thiopental	Tetracycline
	
Accumulates in fat	Accumulates in bone and teeth

3. Metabolism

- Chemical transformation of a drug to enhance elimination.

- In pharmacology, the word metabolism often refers to the process of making a drug more polar and water soluble.

- Sites of biotransformation include liver (main), GI tract, lung, plasma, kidney

Cytochrome P450 System

- The CYPs are a super family of heme proteins that are grouped into families and subfamilies according to their amino acid sequence.

- These proteins are responsible for the metabolism of drugs, chemicals, and other substances

- Drug metabolism by CYP450 takes on two pathways

Phase I (P450) reactions

- Minor molecular changes introduce or unmask polar groups on a parent compound to increase water solubility (e.g. oxidation-reduction, hydrolysis, hydroxylation).

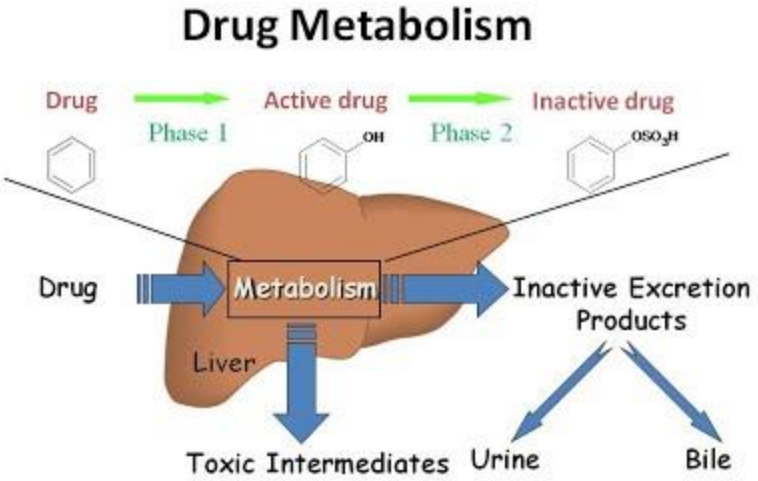
- Product of the reaction can be excreted or undergo further phase II reactions.

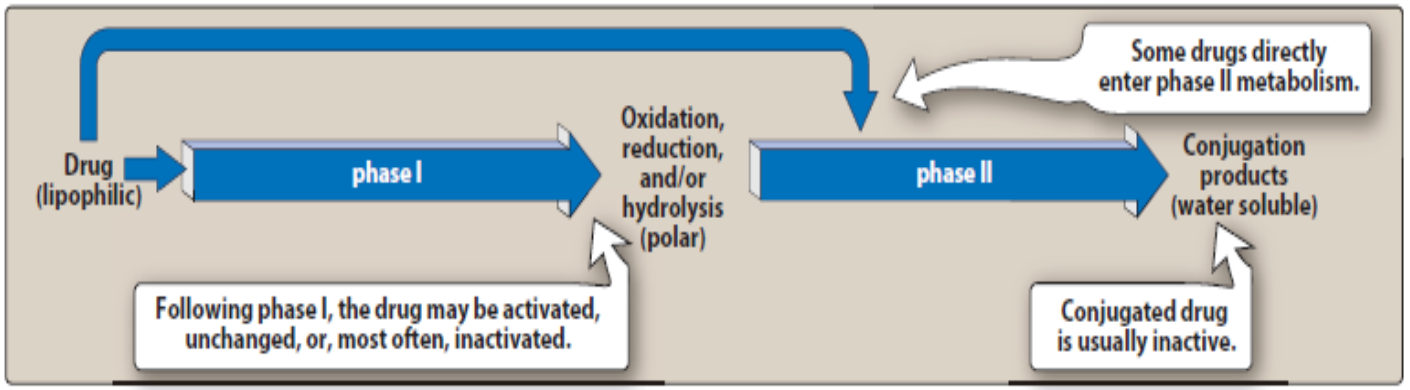
Phase II (conjugation) reactions

- Conjugation with large polar endogenous substrates (e.g. glucuronidation, glutathione conjugation) dramatically increases water solubility and renal elimination.

- Can occur independently of phase I reactions

Phases of metabolism





Factors affecting drug metabolism

- Genetic polymorphisms of metabolizing enzymes
  - Individual genotypes may determine rate of drug metabolism
  - Example – codeine (In Ethiopian patients)

- Enzyme inhibition (due to other drugs)
  - CYP inhibition leads to an increased concentration and bioavailability of the substrate drug
  - E.g. Erythromycin (inhibitor) + simvastatin

    Warfarin + metronidazole (inhibitor)

cont'd …

- Enzyme induction
  - Certain medications enhance gene transcription leading to an increase in the activity of a metabolizing enzyme
  - E.g. Phenobarbital /Carbamazepine+ OCPs

    Rifampicin + OCPs

- Liver dysfunction - hepatitis, alcoholic liver

- Age – neonates and elderly

cont'd …

- Alcohol – acute consumption inhibit CYP but chronic consumption induces CYP450
  - Acetaminophen toxicity (due to toxic metabolites)

- Smoking – induce CYP450 enzymes
  - Theophylline, antipsychotic drugs

4. Excretion

- Transportation of unaltered or altered form of drug out of the body.

- The major processes of excretion include renal excretion, hepatobiliary excretion and pulmonary excretion.

- The minor routes of excretion are saliva, sweat, tears, breast milk, vaginal fluid, nails and hair.

Renal excretion

- The excretion of drug by the kidney involves the following :-
  - Glomerular filtration
  - Active tubular secretion
  - Passive tubular reabsorption

The function of glomerular filtration and active tubular secretion is to remove drug out of the body, while tubular reabsorption tends to retain the drug.

Glomerular filtration

- A passive process, so that only the free drug fraction can be eliminated.

- Drug filtration rate depends on
  - GFR
  - Degree of protein binding of drug
  - Size of drug

Tubular secretion

- An active process that is saturable allowing both protein-bound and free drug fractions to be excreted

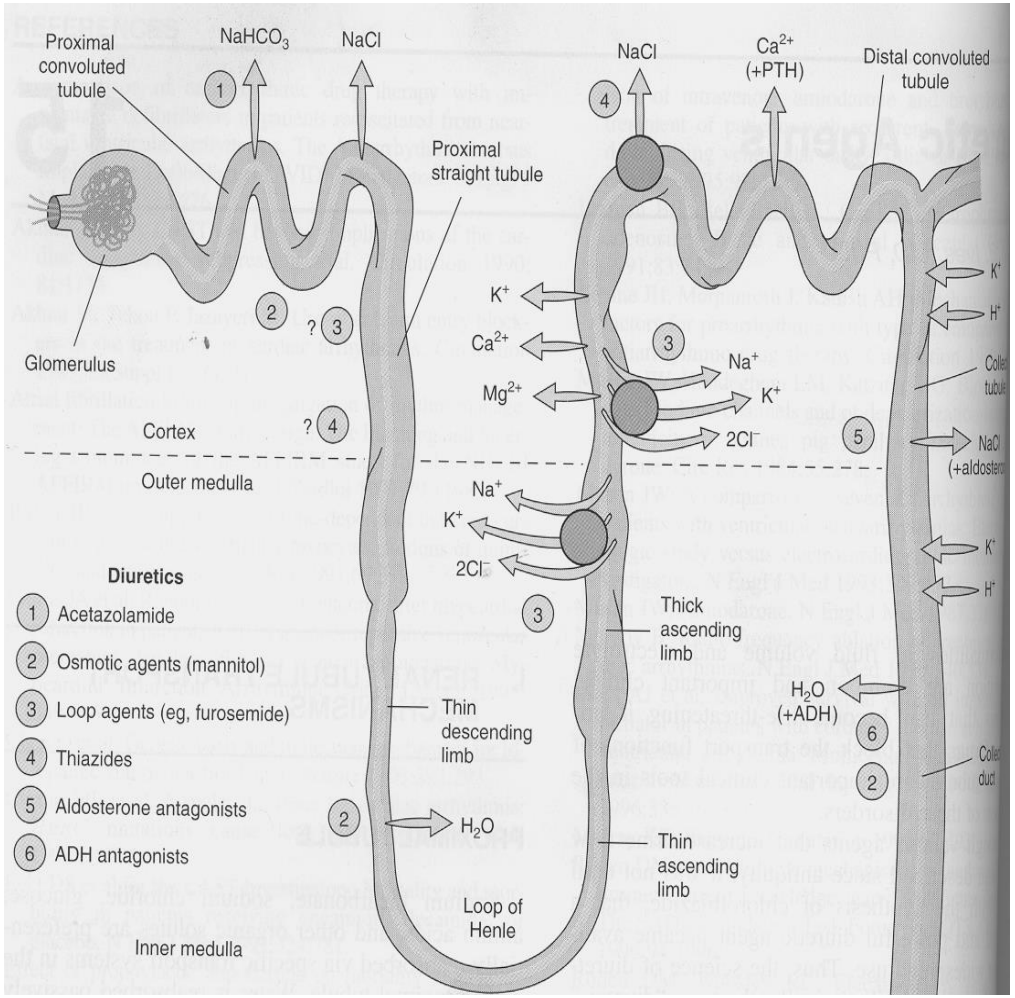
- Drugs may competitively block mutual secretion if both use the same secretion system (e.g. probenecid can reduce the excretion of penicillin)

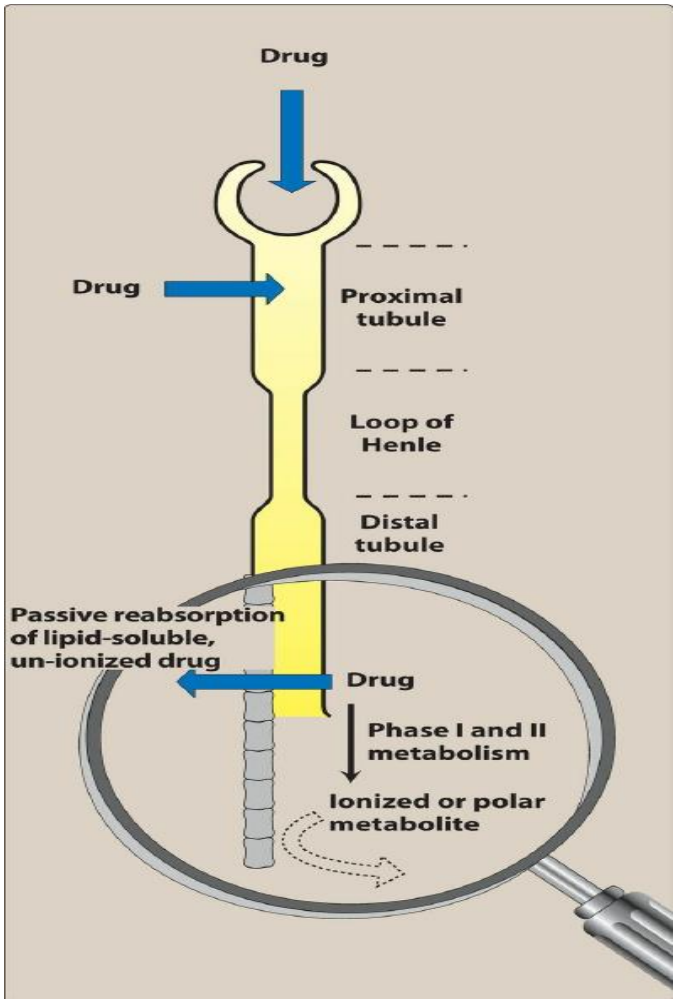
Tubular reabsorption

- Drugs can be passively reabsorbed back to the systemic circulation, countering elimination mechanisms.

- Renal function decreases with age and is affected by many disease states such as diabetes.

cont'd …


Proximal convoluted tubule
$NaHCO_3$
NaCl
Proximal straight tubule
NaCl
Ca^{2+} (+PTH)
Distal convoluted tubule
Glomerulus
Cortex
Outer medulla
Inner medulla
Thick ascending limb
Thin descending limb
Thin ascending limb
Loop of Henle
H_2O (+ADH)
H_2O
Diuretics
1 Acetazolamide
2 Osmotic agents (mannitol)
3 Loop agents (eg, furosemide)
4 Thiazides
5 Aldosterone antagonists
6 ADH antagonists


Drug
Drug
Proximal tubule
Loop of Henle
Distal tubule
Passive reabsorption of lipid-soluble, un-ionized drug
Drug
Phase I and II metabolism
Ionized or polar metabolite

Excretion via other routes

- Some drugs and metabolites are actively excreted in the bile or directly into the GI tract
- Enterohepatic reabsorption counteracts stool elimination, and can prolong the drug's duration in the body
- Some glucuronic acid conjugates that are excreted in bile may be hydrolyzed in the intestines by bacteria back to their original form and can be systemically reabsorbed.

cont'd …

- Lungs - elimination of anesthetic gases and vapors by exhalation.
  - Most of the alcohol is excreted by lungs

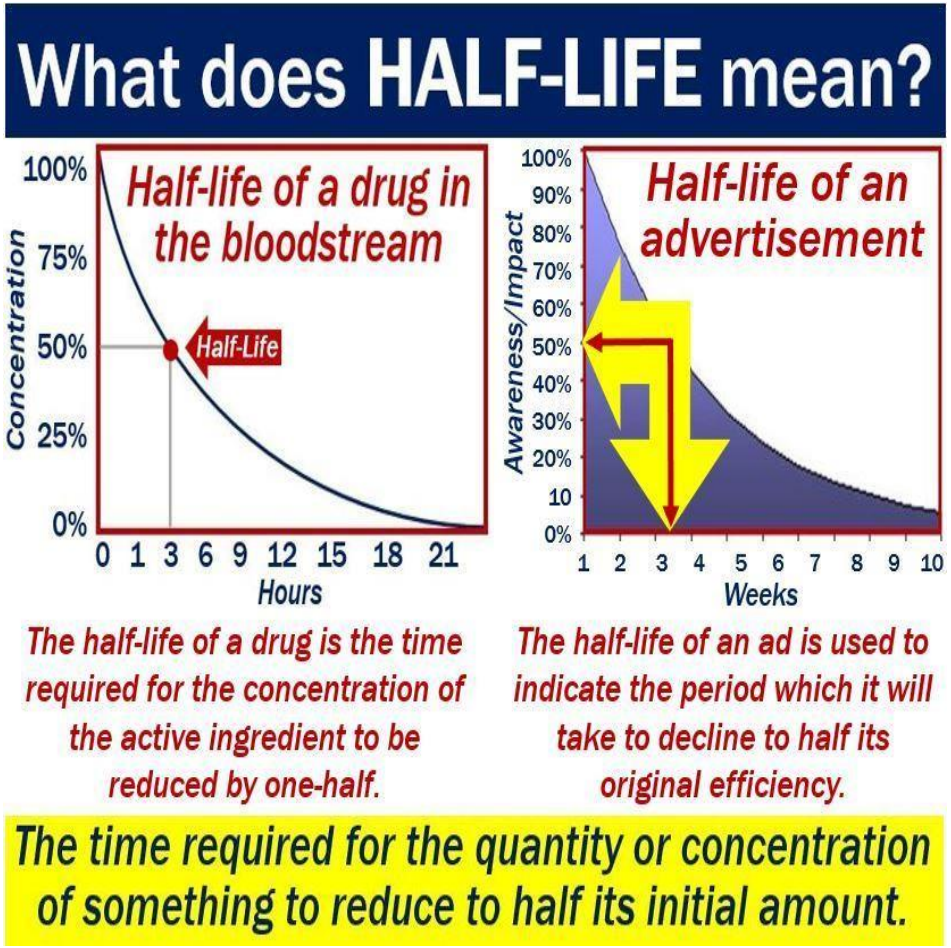
- Saliva - saliva concentrations of some drugs parallel their plasma levels
  - Example – rifampin

Factors affecting excretion

- Renal disease
  - Results in decreased drug clearance

Half-Life ($t_{1/2}$)

- Time taken for the serum drug level to fall 50% during elimination.



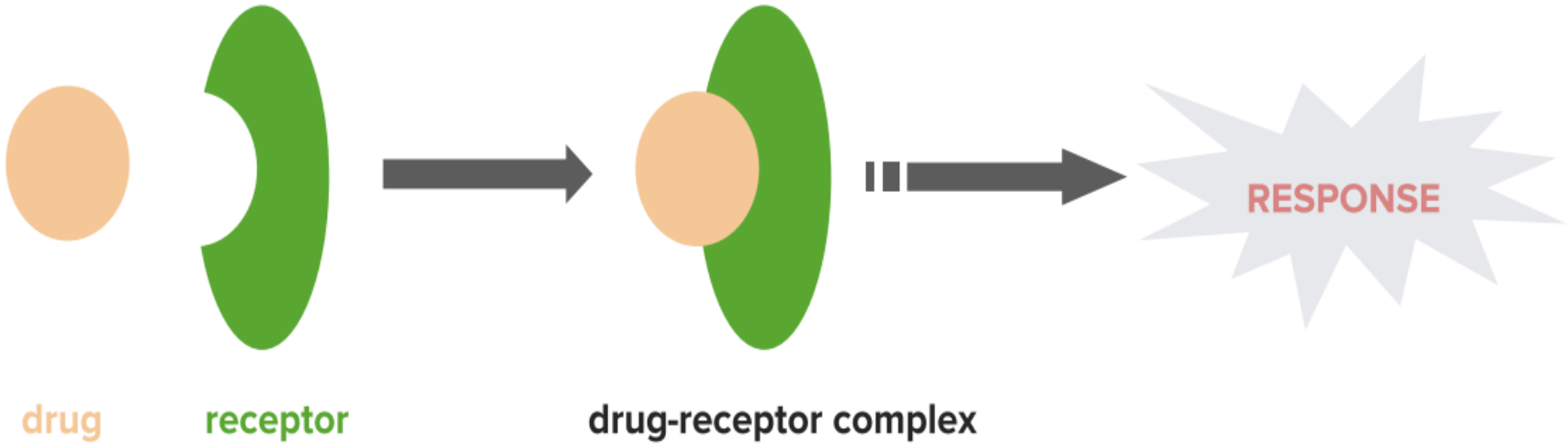
Pharmacodynamics

- Describes the actions of a drug on the body and the influence of drug concentrations on the magnitude of the response.

- Most drugs exert their effects, both beneficial and harmful, by interacting with receptors present on the cell surface or within the cell.

Receptors

- Component of a cell or organism that interacts with a drug and initiates the chain of event leading to drug's observed effects.



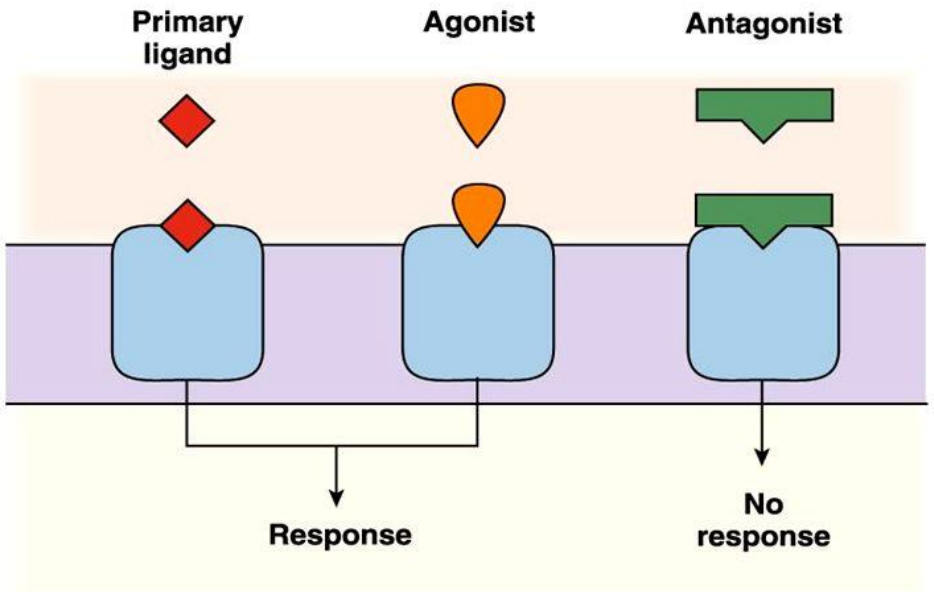
Agonist and antagonist

- Agonist
  - Binds to a receptor and activates the receptor to produce a biological response.

- Antagonist (anti - agonist)
  - Blocks a biological response by binding to and blocking a receptor from binding to endogenous neurotransmitters.

Agonist vs. antagonist

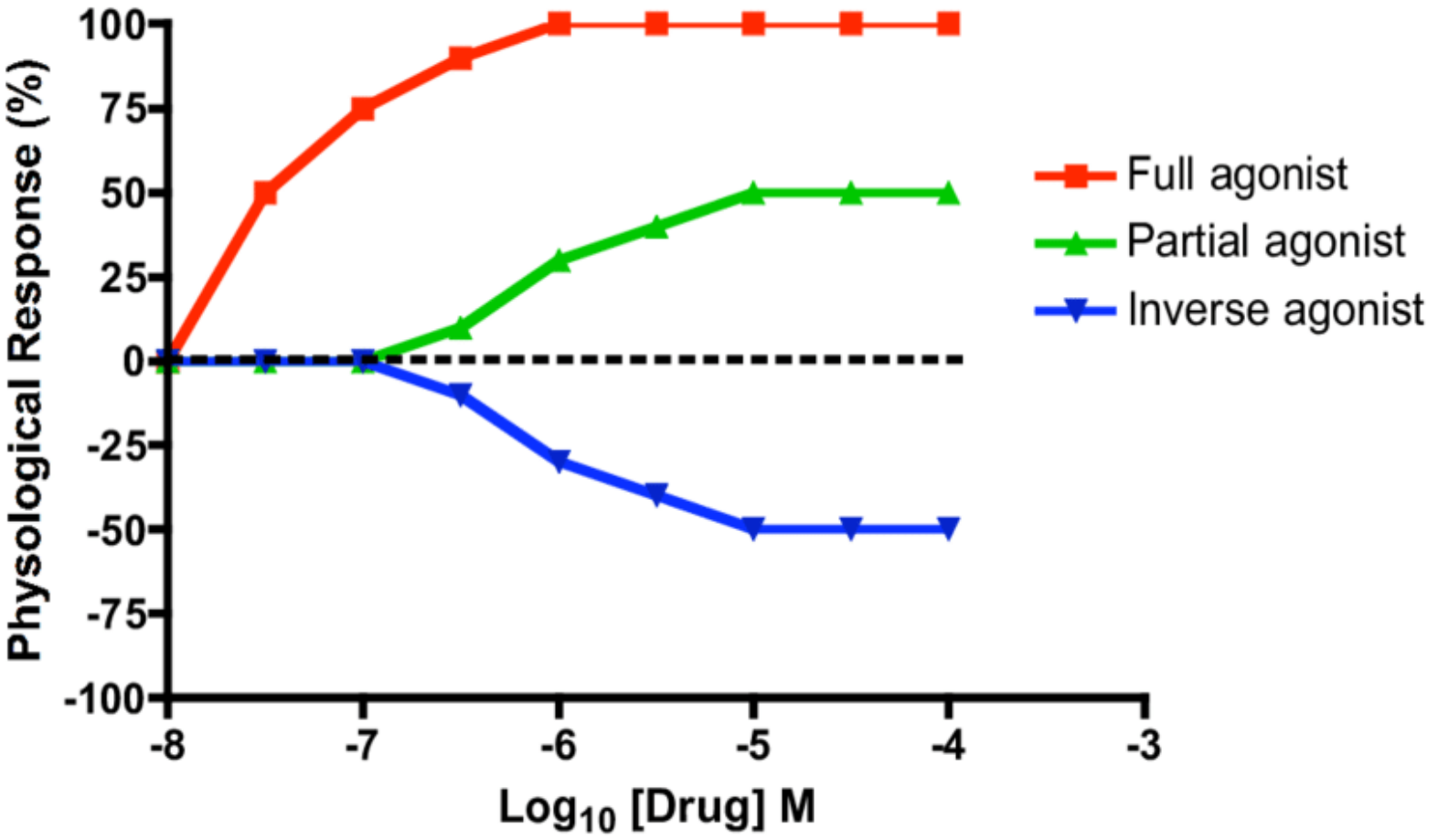
An antagonist occupies but does not activate the drug receptor



Types of agonist

- Full agonists
  - Elicit a maximal effect at a receptor
  - E.g. phenylephrine at α1-adrenoceptors
- Partial agonists
  - Only elicit a partial effect, no matter the concentration at the receptor.
  - E.g. aripiprazole (atypical antipsychotic) at DA receptors
- Inverse agonists
  - Stabilize the inactive receptor state
  - Convert R active to R inactive, which is activated intrinsically.

cont'd …


Physiological Response (%)
100
75
50
25
0
-25
-50
-75
-100
-8
-7
-6
-5
-4
-3
Log10 [Drug] M
Full agonist
Partial agonist
Inverse agonist

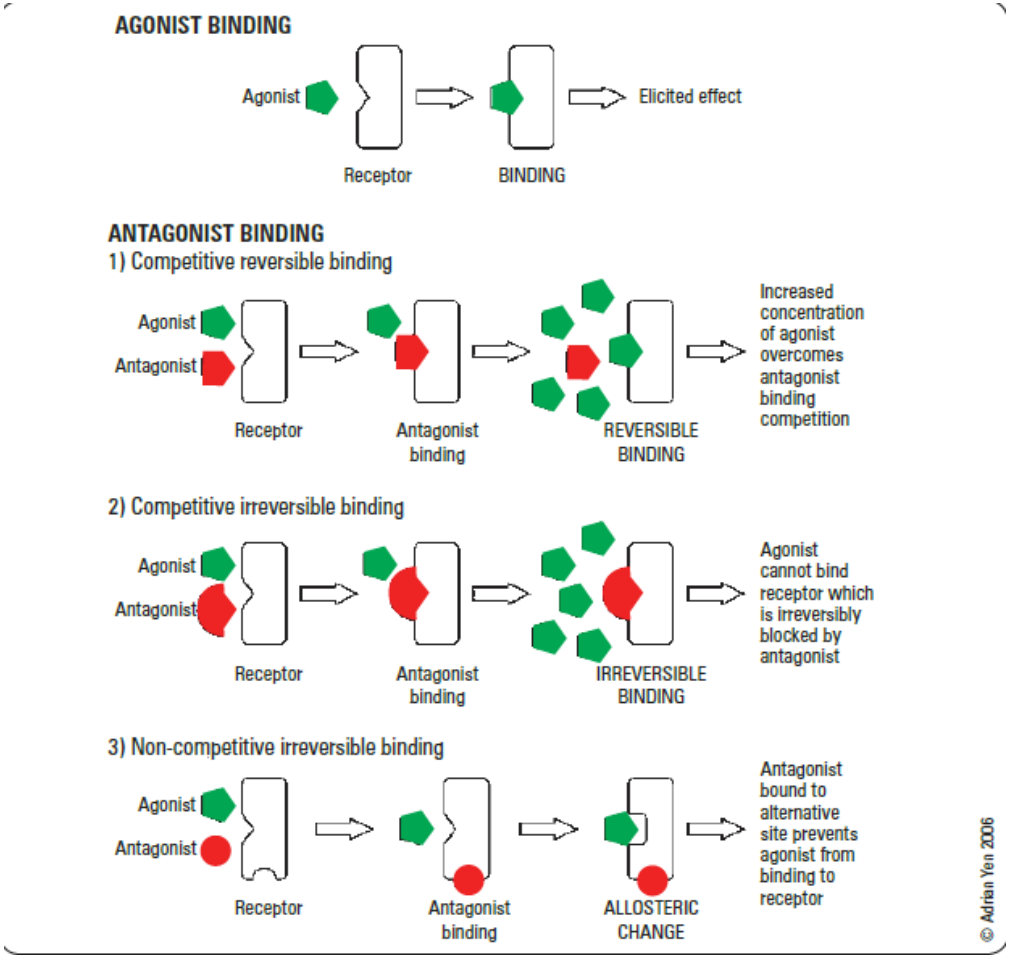
Antagonism

- Competitive antagonists
  - Reversible
    - Reversibly bind to the same receptor as the agonist, thus displacing it (e.g. naloxone is an antagonist to morphine or heroin)
  - Irreversible
    - Form a covalent bond with the receptor, thus irreversibly blocking substrates from binding (e.g. phenoxybenzamine forms a covalent bond with adrenergic receptors preventing adrenaline and NE from binding)

cont'd …

- Non-competitive antagonism
  - Antagonist binds to an alternate site near the agonist site, producing allosteric effects that change the ability of the agonist to bind
  - E.g. organophosphates irreversibly bind acetylcholinesterase

cont'd …


AGONIST BINDING
Agonist
Elicited effect
Receptor
BINDING
ANTAGONIST BINDING
1) Competitive reversible binding
Agonist
Antagonist
Receptor
Antagonist binding
REVERSIBLE BINDING
Increased concentration of agonist overcomes antagonist binding competition
2) Competitive irreversible binding
Agonist
Antagonist
Receptor
Antagonist binding
IRREVERSIBLE BINDING
Agonist cannot bind receptor which is irreversibly blocked by antagonist
3) Non-competitive irreversible binding
Agonist
Antagonist
Receptor
Antagonist binding
ALLOSTERIC CHANGE
Antagonist bound to alternative site prevents agonist from binding to receptor
© Adrian Yen 2006

Dose response relationships

- Graded dose effect (Individual level)
  - As the dose administered to a single subject or tissue increases, the pharmacological response also increases in graded fashion up to ceiling effect
  - Potency, efficacy

- Quantal dose effect (population level)
  - It is all or none response, sensitive objects give response to small doses of a drug while some will be resistant & need very large doses.
  - Safety (LD50), effectiveness(ED50)

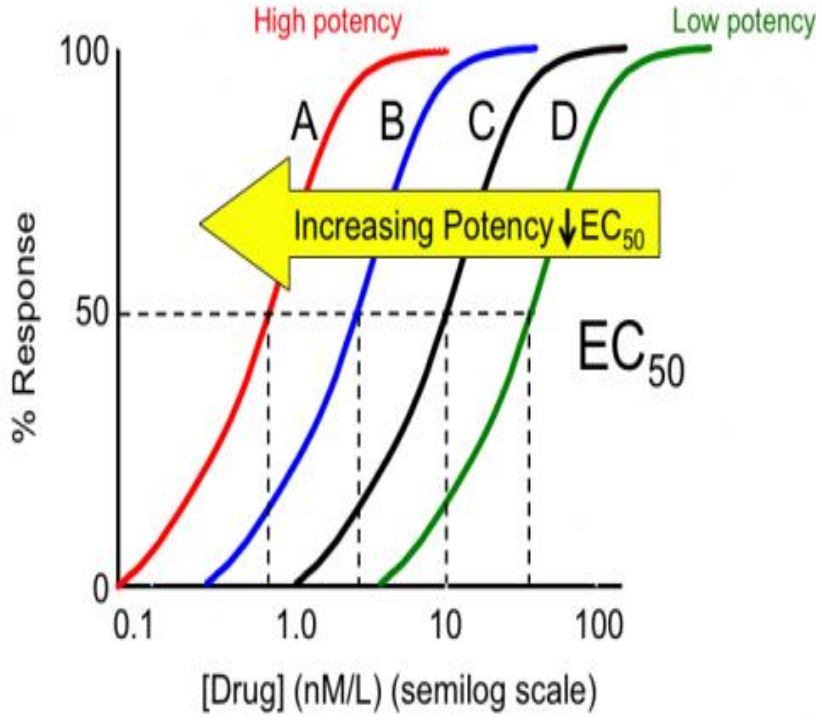
Graded dose effect

- ✓ Potency
  - The amount of drug required to produce 50% of the maximal response that the drug is capable of inducing.
  - Candesartan (4 to 32mg) and irbesartan (75 to 300mg)

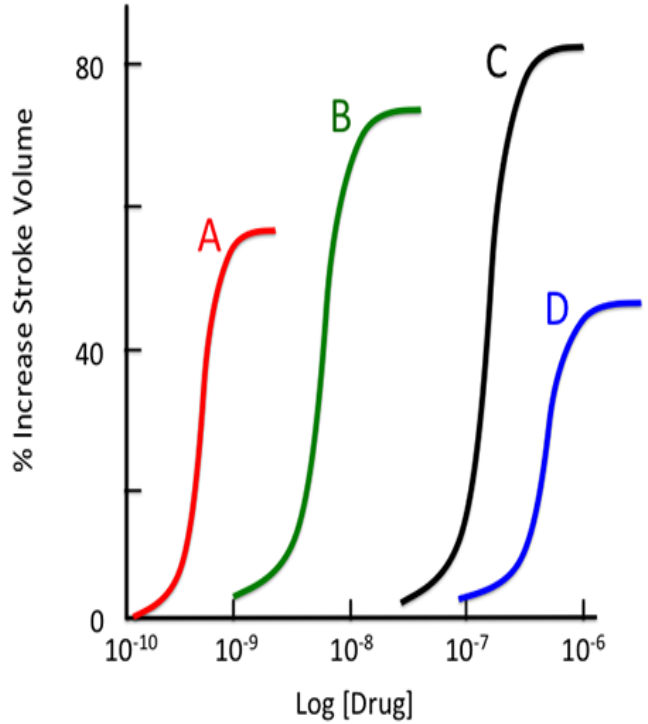
- ✓ Efficacy
  - Measures the maximal effect of a drug independently of the dose (concentration).

Potency and efficacy

Potency



Efficacy



Quantal dose effect

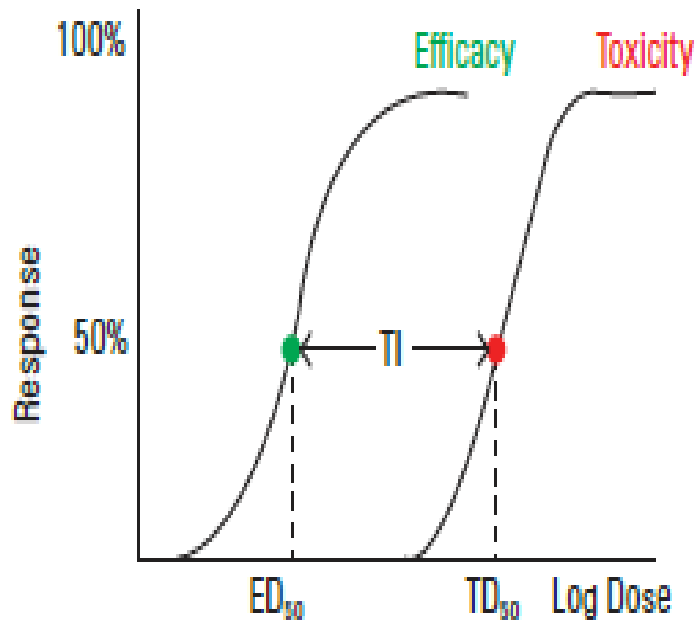
- Safety
  - Toxic dose (TD50): the dose, which would be toxic for 50% of the test population.

- Effectiveness
  - Effective dose (ED50): the dose, which produces a desired therapeutic effect in 50% of test population.

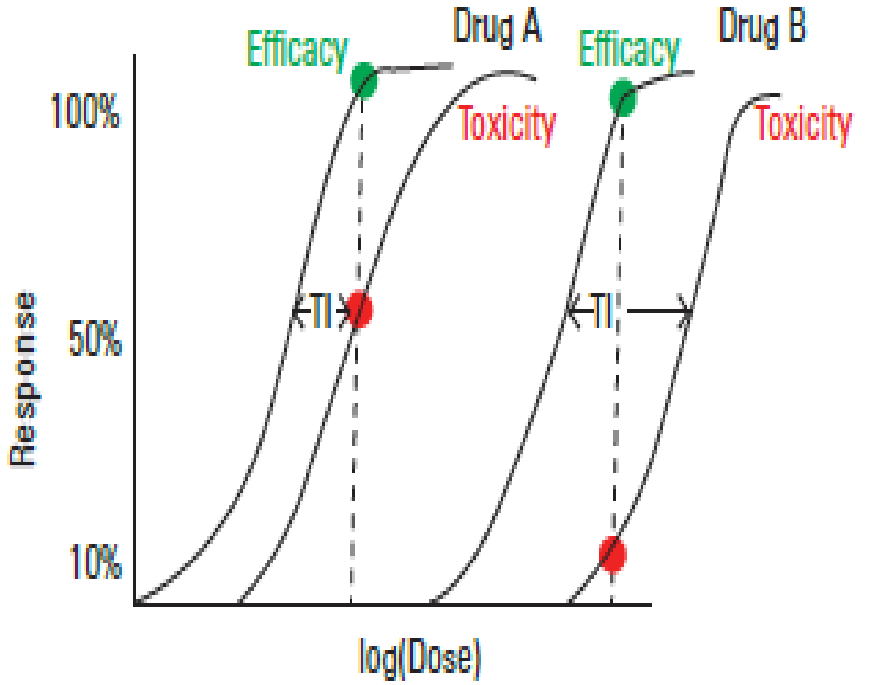
Therapeutic index (window)

- Reflects the “margin of safety” for a drug
- T.I = TD50/ED50
- The larger the T.I, the safer the drug
  - Warfarin – narrow T.I (unsafe drug)
  - Penicillin – wide T.I (safe drug)

T.I … cont'd



The therapeutic index (TD_{50}/ED_{50}) is a measure of the margin of safety of a given drug



Drug A has a much narrower therapeutic index than Drug B. The dose of Drug A required to achieve a 100% therapeutic response will be toxic in 50% of patients. For Drug B, this is only 10%

Drug interaction

- Harmful interaction
  - Diazepam + IV fluids = loss of effect
  - Tetracycline + milk = decreased absorption
  - Rifampicin + OC = pregnancy
  - Warfarin + metronidazole = bleeding

Beneficial D/I

Table 1.4 Types of Drug Interactions

TYPE OF INTERACTION	MATHEMATICAL MODEL
Addition - The response elicited by combined drugs is EQUAL TO the combined responses of the individual drugs.	1 + 1 = 2
Synergism - The response elicited by combined drugs is GREATER THAN the combined responses of the individual drugs.	1 + 1 = 3
Potentiation - A drug which has no effect enhances the effect of a second drug.	0 + 1 = 2
Antagonism - Drug inhibits the effect of another drug. Usually, the antagonist has no inherent activity.	1 + 1 = 0

Adverse drug reactions

1. Side effects
2. Untoward effects
3. Allergic reactions
4. Teratogenic effects.

Side effects

- Pharmacological effects produced with therapeutic dose of the drug.
- Side effects can be beneficial or harmful.
- Examples include atropine and antihistamine drugs.

Untoward effect

- Develop with therapeutic dose of a drug.
- They are undesirable and if very severe, may necessitate the cessation of treatment.
- Examples
  - Diarrhea with ampicillin
  - potassium loss with diuretics

Allergic reactions

- Most of the drugs used in therapeutics are capable of causing allergic or hypersensitive reactions.
- These reactions may be mild or very severe like anaphylaxis.
- When an individual has been sensitized to an antigen (allergen) further contact with that antigen can sometimes lead to tissue damaging reactions.

Allergy to antibiotics



Teratogenic effect

- Some drugs given in the first three months of pregnancy may cause congenital abnormalities and are said to be teratogenic.

- The best known example is thalidomide which results in early easily recognizable abnormalities such as absent or grossly abnormal limbs.

Thalidomide incident





FDA Pregnancy Categories

Category	Interpretation
A	Controlled studies show no risk: Adequate, well-controlled studies in pregnant women have failed to demonstrate risk to the fetus.
B	No evidence of risk in humans: Either animal findings show risk, but human findings do not; or, if no adequate human studies have been done, animal findings are negative.
C	Risk cannot be ruled out: Human studies are lacking, and animal studies are either positive for fetal risk or lacking as well. However, potential benefits may justify potential risk.
D	Positive evidence of risk: Investigational or postmarketing data show risk to the fetus. Nevertheless, potential benefits may outweigh risks.
X	Contraindicated in pregnancy: Studies in animals or humans, or investigational or postmarketing reports, have shown fetal risk that clearly outweighs any possible benefit to the patient.

THE END

THANK YOU